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Evaluating Latent Profile Analysis for Persona Development in Human-Computer Interaction

by

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Abstract

User personas are widely used in Human-Computer Interaction (HCI) to support user-centred decision making.

Data-driven approaches to user persona development within Human-Computer Interaction (HCI) and User Experience (UX) have widely relied on hard-clustering techniques to create user groupings. However, these hard-clustering methods produce mutually exclusive groupings that may oversimplify behavioural diversity.

This thesis investigates the potential of a probabilistic clustering method, Latent Profile Analysis (LPA), as an alternative approach for behavioural segmentation in persona development. By comparing and analysing three behavioural datasets with k-means clustering, agglomerative hierarchical clustering, and LPA, this project aims to assess the cluster composition, behavioural differentiation, and interpretability of the resulting cluster structures.

The overall findings suggest that LPA identifies more granular behavioural patterns; however, the additional clusters do not always translate into more practical or interpretable personas. This highlights the trade-off between simplicity, behavioural granularity and usability that's necessary for successful user-persona development.

This research contributes to ongoing studies in data-driven UX design by providing an empirical comparison of clustering approaches and by evaluating their practical usefulness for persona creation.

Abbreviations

LPA	Latent Profile Analysis
HCI	Human-Computer Interaction
UX	User Experience
SSE	Sum of Squared Errors

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Chapter 1

1. Introduction

User personas are a core tool in Human-Computer Interaction (HCI) and User Experience (UX) design, not only supporting designers in comprehending user needs but also enabling them to understand user behaviours and goals throughout the product development lifecycle (Wang et al., 2025; Hudson, 2025). With the aim of personas being to successfully and accurately describe real people and user groupings, it also showcases an enormous obstacle in defining from data using traditional, rigid clustering approaches (Holverstott, 2021). This suggests that current clustering techniques in user persona creation methods may fail to account for and capture the complexity of real-world user behaviour.

Data-driven approaches to user persona development are well established in HCI, particularly traditional clustering methods such as k-means and hierarchical clustering (Salminen et al., 2021). These traditional clustering methods are heavily relied on due to their simplicity, accessibility, and long-standing reliability (Kent et al., 2014). However, these techniques equally rely on rigid assumptions that may oversimplify behavioural patterns, limiting their ability to capture nuanced differences among users (Liu et al., 2022). As data collection and user interaction data become more complex and rich, there is a clear growing need for segmentation methods that are more flexible and interpretable, and can adequately communicate the behavioural and psychological patterns favourable for persona creation.

This thesis investigates Latent Profile Analysis (LPA), a probabilistic clustering technique commonly utilised in psychology and behavioural sciences, as an alternative method for persona generation in UX design and HCI (Gilroy et al., 2022). This study uniquely contributes through its systematic comparison of LPA with established clustering methods, such as k-means and agglomerative

hierarchical clustering, to assess their potential to produce user clusters that better reflect behavioural complexity, interpretability, and practical relevance for persona creation. By employing all three methods on selected datasets and evaluating them through both quantitative and qualitative criteria, including behavioural richness, interpretability, model performance, and utility for persona development, this thesis provides new insights into the potential advantages of model-based clustering for user persona generation. By examining both technical performance and practical applicability of these methods, this research seeks to clarify whether LPA addresses important shortcomings of traditional techniques and offers a significant step forward in supporting effective, data-driven persona creation in UX design.

The remainder of this thesis is structured as follows. Chapter 2 reviews the literature on persona development, clustering techniques, and applications of LPA in HCI. Chapter 3 outlines the study's methodology and evaluation framework. Chapter 4 addresses unique steps in implementation. Chapter 5 describes and evaluates project results and findings. The thesis then concludes with a discussion of current progress, limitations, and future research avenues.

Chapter 2

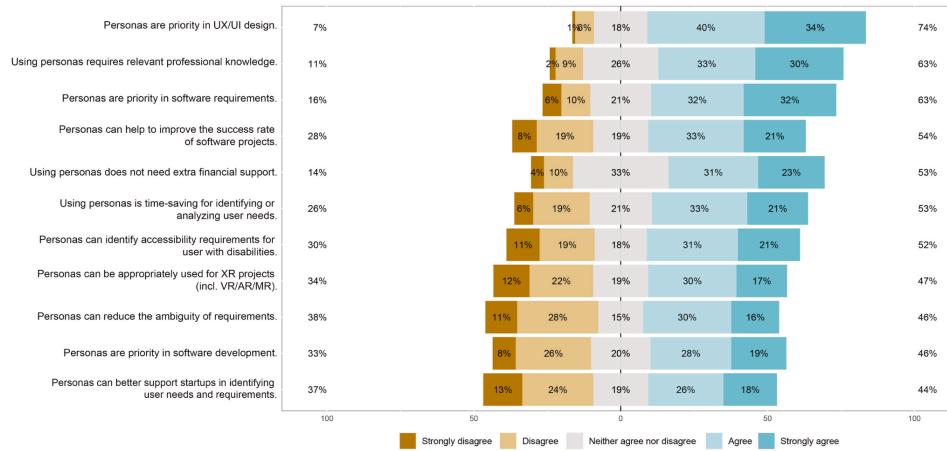
2. Background

2.1 Personas in UX and HCI

User personas are central to Human-Computer Interaction (HCI) and user experience (UX) design. These fictional, data-driven representations of real users help design teams create products that meet real user needs in simple, intuitive, and efficient ways (Blomkvist, 2002; Wang et al., 2025). By taking the abstract idea of a user, and giving them a name, face and story, user personas help product teams, ranging from developers, to designers and engineers, have a collective, data-backed, user-centred understanding of their stakeholders. These “archetypical representations of real or potential users” are intended to help product teams stay grounded in user needs by designing with empathy, keeping user goals and frustrations in mind, and ideally minimising bias (Blomkvist, 2002).

User personas are especially important at the early stages of the design process, where they help align teams and clarify requirements. As shown by the survey data in *Figure 2.1*, personas are highly rated as a priority tool in UI/UX design (Wang et al., 2024) and across multiple stages of product development where they have been recorded to support organisation, conceptualization, ideation, prototyping, education, copywriting, prioritization and communication (Salminen et al., 2022).

Figure 2.1: Scaled rating of persona usage by respondent perspectives (Wang et al., 2024).



The primary benefit of user personas is to help bridge the gap between user research and skillful design decisions, ideally creating a shared mental model that brings structure to team discussions and establishes a consistent language about user types across disciplines (Siika-aho, 2023). This is particularly crucial in multidisciplinary projects prevalent in UX and HCI where improving collaboration, supporting clear prioritisation, and reducing scope creep is priority (Salminen et al., 2022).

Effective, well-produced user personas help:

- Improve communication between cross-functional teams,
- Foster empathy for real user challenges,
- Ensure product decisions are user-centred,
- Help prioritise and scope features based on real user needs, and
- Surface and challenge the team's implicit assumptions.

Despite their usefulness, personas face the major concern of oversimplifying user diversity and behaviour. Generic persona development for large groups of potential users is very likely to fail to represent its users adequately by simplifying “the role definition as a set of expected behaviours” and “masks the complexity and ambiguity that is discovered” (Hudson, 2013). This limitation is further reinforced by UX professionals who reported personas are often perceived as highly subjective due to their reliance on interview-based data and limited use of rigorous empirical-grounded analysis methods. Coupled with practical team concerns involving a lack of expertise with data analysis skills like statistics and psychology, there is the struggle to consistently find the right level of abstraction (Wang et al., 2024).

These findings suggest that while personas are acknowledged as powerful tools in HCI and UX/UI design, they are often inconsistently or incorrectly applied, under-resourced, and under-taught. Overall, mutually highlighting the importance of user personas as they anchor design decisions in real-world contexts whilst establishing the need for more flexible, statistically grounded and inclusive persona practices that take into account the complex nature of users whilst being highly adaptable to data sources and balance user diversity.

2.2 Data-driven Persona Development Methods

Traditionally, user persona development follows a cycle of user research, data analysis, and persona development. Persona development methods can be categorised into three main categories pertaining to qualitative, quantitative, and hybrid approaches.

Firstly, we have personas created through qualitative methods such as interviews, focus groups, and ethnographic research. This is often the preferred method for creating personas due to its simplicity and ability to collect data rich in context surrounding user motivation and behaviours; however, these methods can be time-consuming, expensive to execute and scale, and data can be inadvertently biased upon interpretation and application (Siika-aho, 2023).

Then there are quantitative approaches which leverage large datasets from analytics, surveys, or public sources to identify behavioural and demographic patterns, using statistical models or machine learning to segment users (Patkar et al., 2023). While promising in terms of scalability and repeatability, many such approaches rely on hard clustering techniques that assign users to a single group, failing to reflect overlap or ambiguity in real behaviour.

Hybrid approaches aim to balance the strengths of both quantitative and qualitative methods by combining rich behavioural insights with measurable data attributes to create a more comprehensive picture of the user (Liang, 2020). However, these existing data-driven approaches still rely on rigid clustering techniques which limits their ability to fully capture the behavioural complexity and nuance inherent in real user behaviour.

And more recently, HCI has seen a growing interest in behaviour-driven personas, which focuses on building user profiles based on how users interact with systems, what features they use, and their patterns of engagement rather than relying on surface level demographic characteristics (Jain, 2025). By focusing on usage patterns and insights derived from available system analytics, behaviour-driven

personas draw a shift in UX/UI and HCI towards system-centred understanding of audiences. Hence, establishing user behaviour as a more reliable basis that can quickly inform UI design decisions (Medium, 2016).

2.3 Clustering Methods for Persona Development

Clustering, also known as cluster analysis, is an unsupervised machine learning technique used to group similar data points in “clusters” that share the greatest similarity. Data points are grouped together based on attributes like outcomes or data labels (Gao et al., 2023). In this report, traditional clustering techniques refer to k-means and hierarchical clustering as they represent common, well-documented methods in the literature and will be used to provide a solid baseline for assessing the potential value of LPA in user persona development.

Traditional methods are valued for their simplicity, scalability, and interpretability, but also have limitations, such as sensitivity to input parameters, susceptibility to outliers, and the inability to represent overlapping memberships (Quan et al., 2015).

Overall, clustering methods can typically fall into either of these four categories:

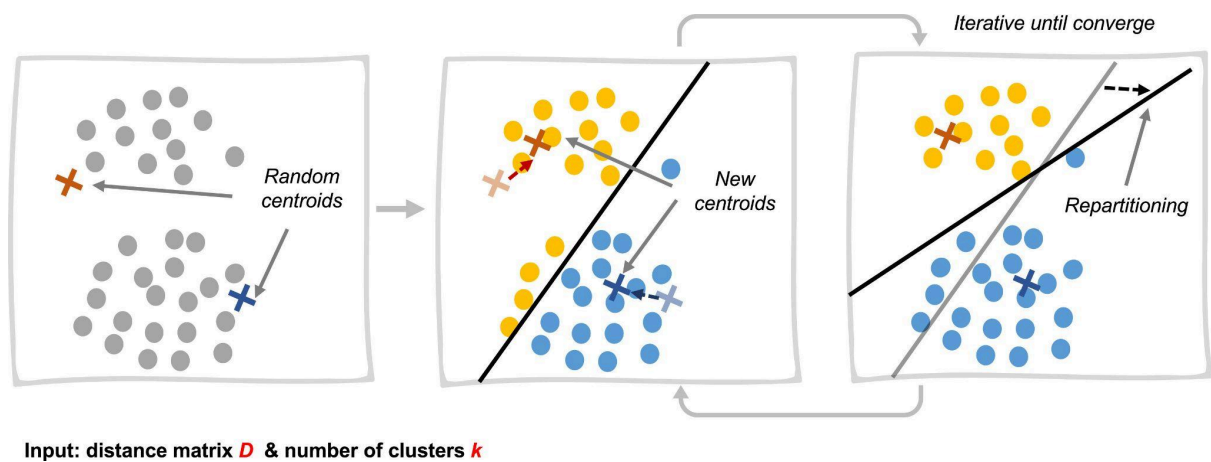
- Centre-based partitioning clustering,
- Hierarchical clustering,
- Density-based clustering, and
- Model-based clustering

This thesis will focus on three clustering techniques: k-means clustering for centre-based partitioning, agglomerative hierarchical clustering as a hierarchical method, and Latent Profile Analysis (LPA) as a model-based approach. With LPA clustering potentially being a more promising method that enables researchers to identify latent user groups based on behavioural, demographic, or psychographic data.

2.3.1 K-means Clustering

K-means clustering falls under the family of centre-based partitioning models which aim to “find mutually exclusive clusters with spherical shapes based on data points’ distance to cluster centres” (Gao et al., 2023). As seen on *Figure 2.3*, the process begins by randomly placing k cluster centres, known as centroids. Then, each data point is assigned to the nearest centroid, typically using Euclidean distance, and the centroids are continuously updated to represent the mean position of their assigned data points, which continues iteratively until the centroids stabilise.

Figure 2.3: Illustration of K-means Clustering (Gao et al., 2023).

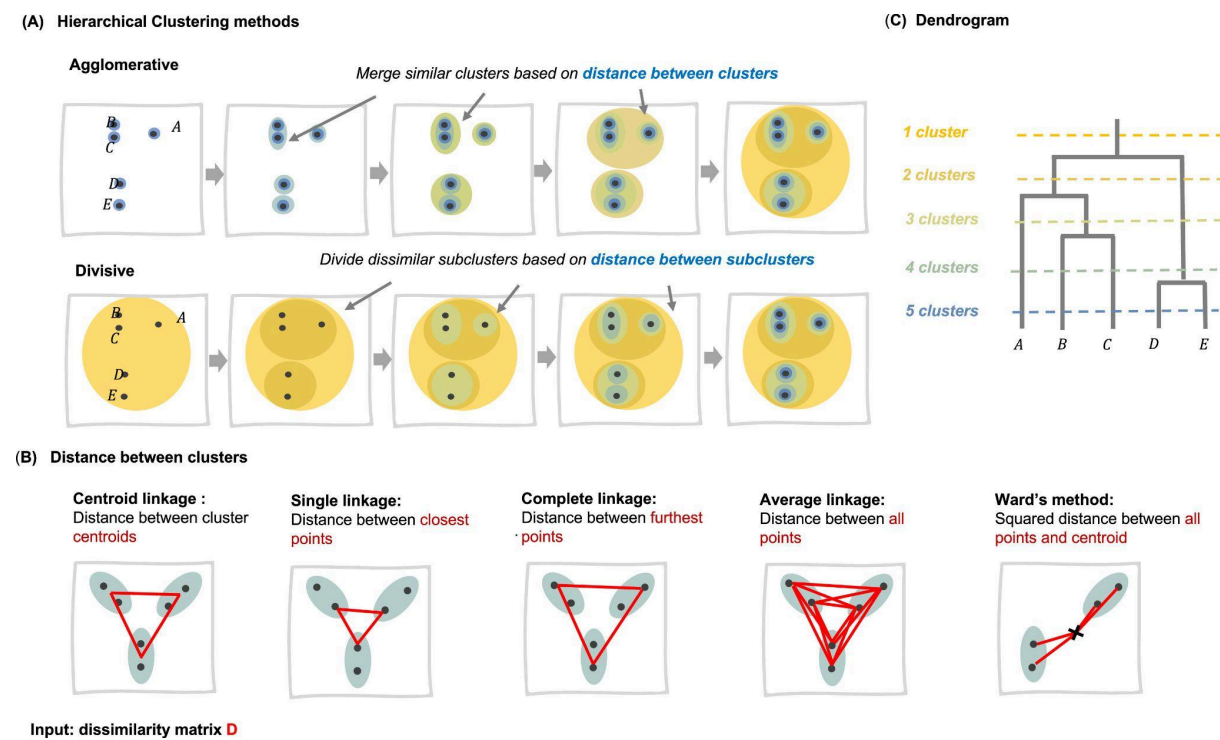


In UX, HCI, and psychology, centre-based clustering methods such as k-means are the most commonly used due to their simplicity, high computational speed, and ease of interpretation. However, this method requires the number of clusters, ‘ k ’, to be specified in advance, cannot identify non-spherical shapes, and is sensitive to outliers and to initial centroid placement (Gao et al., 2023). These factors can limit its effectiveness when working with noisy or complex user data (Liu et al., 2022).

2.3.2 Hierarchical Clustering

Hierarchical clustering is a method that builds a hierarchy of clusters by either progressively merging smaller clusters into larger ones, which is known as the agglomerative approach, or by splitting larger clusters into smaller ones, also known as the divisive approach) (Gao et al., 2023). The agglomerative method begins with each data point as its own cluster. At each step, the two closest clusters are merged based on a distance metric (e.g., Euclidean distance) and linkage criterion (e.g., single, complete, or average linkage). This process continues until all points are contained within a single large cluster. At the end, the results are presented as a dendrogram tree, which allows users the ability to cut at different levels to receive the desired number of clusters. This step-by-step process can be seen in *Figure 2.4*.

Figure 2.4: Illustration of Hierarchical Clustering (Gao et al., 2023).



One of the main advantages of hierarchical clustering is that it does not require a predefined number of clusters to be specified in advance. This flexibility makes it useful for exploratory analysis and aids in visualising the relationships between data points. Unfortunately, hierarchical methods can also come with their own drawbacks, such as being computationally expensive for large datasets and being sensitive to outliers, which may distort hierarchy results.

2.4 Latent Profile Analysis (LPA)

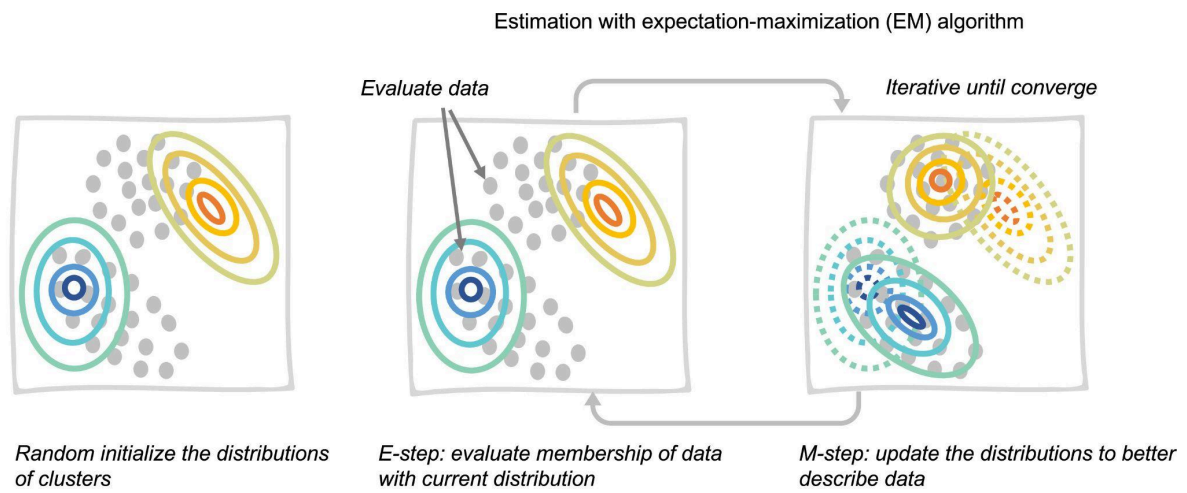
Latent Profile Analysis (LPA) is a model-based clustering technique that identifies latent subgroups (hidden profiles) in the population by modelling patterns in continuous variables (Gao et al., 2023). Unlike distance-based clustering methods, LPA does not depend on the proximity of data points in space. Instead, LPA adopts a probabilistic approach that estimates the likelihood of each participant belonging to each profile by assuming the data is generated from a finite mixture probability distribution. With each distribution corresponding to a latent profile with its own mean pattern and variance structure (Spurk et al., 2020). If the observed variables have normal distributions, then the finite mixture model is classified as a Gaussian Mixture Model (GMM). For continuous data, GMM-based clustering is applied as Latent Profile Analysis (LPA), and conversely, for discrete variables, the model becomes Latent Class Analysis (LCA) (Gao et al., 2023).

To summarise, the LPA model starts by randomly initialising parameters on every cluster and iteratively refines them, aiming to align current distributions to match observed data. Lastly, the Expectation-Maximisation (EM) algorithm is applied continuously until the parameters converge (Gao et al., 2023). As shown in *Figure 2.5*, the EM algorithm follows a two-step process involving an E-step and an M-step:

- E-step: Estimates the probability (weight) that each data point belongs to each cluster.
- M-step: Adjusts the distribution's parameters based on the E-step.

Additionally, the optimal number of profiles is not pre-determined but is guided by model selection criteria such as the Bayesian Information Criterion (BIC) or Akaike Information Criterion (AIC), along with diagnostics like entropy to assess the classification precision (Spurk et al., 2020).

Figure 2.5: Illustration of Finite Mixture Model (Gao et al., 2023).



Input: **data** , type of **distributions** and number of clusters **k**

2.4.1 Latent Profile Analysis in UX design and HCI

LPA is especially well suited for survey and Likert-scale data, which is common in UX research (Williams et al., 2016). This allows researchers to use LPA to identify subtle variations and differences in attitudes, behaviours, or preferences that traditional clustering tends to overlook (Gao et al., 2023). Based on this overview, LPA shows great promise for data-driven, behaviour-based user persona creation in HCI and UX design.

Whilst existing applications of k-means in persona development show that it can effectively segment large-scale datasets into distinct user groups with clear demographic and interest-based profiles, it also revealed that reliance on purely quantitative features neglects deeper behavioural modelling and may oversimplify user diversity within clusters which limits the understanding of contextual data required in UX research (Quan et al., 2020; Liu et al., 2022).

These limitations present an avenue where LPA can potentially be established as a reliable method focused on behavioral persona development that bridges the gap between the simplicity and reliability of traditional clustering methods and the need for more behaviourally nuanced user segmentations. So far, promising results in research studies comparing LPA and k-means consistently concluded LPA displays more insightful latent profiles not captured through distance-based approaches (Liu et al, 2022). Additionally, existing work in vocational behaviour outline best practices for applying LPA to behavioural segmentation where LPA supports evidence-based design by connecting psychological traits to observable patterns (Spurk et al. 2020).

This makes LPA highly promising for behavior-based persona creation in UX and HCI where its ability to handle subjective data, expose latent motivations, and support probabilistic interpretation provide a statistically grounded lens in the identification of meaningful user differences providing insights not just in what people do, but why they do it.

2.5 Research Gap and Motivation

Despite growing interest in data-driven persona creation, the majority of segmentation techniques used in UX and HCI research continue to rely on traditional clustering methods. While these approaches are valued for their simplicity and accessibility, they often lack the statistical robustness required to uncover deeper behavioural patterns which results in limits on the interpretability and practical usefulness of the resulting personas.

Model-based clustering methods, particularly Latent Profile Analysis (LPA), have shown promise during direct comparisons to traditional clustering methods in behavioural research, however, unlike k-means, LPA has not yet been directly applied to persona development in UX. The success of LPA in identifying psychological and behavioural profiles builds a strong foundation for exploring its application to persona development, where understanding diverse, complex psychological and behavioural indicators in user needs potentially offers more individualised and interpretable profile segmentations (Yazdi et al., 2025). This illuminates the gap in literature where the practical value of LPA for guiding UX design decisions remains untested. The ample precedent of comparative evaluations between LPA and traditional clustering methods in psychology research has provided valuable methodological information, but to date, there is no research directly comparing LPA to traditional clustering methods for persona development in UX contexts.

This study addresses these gaps by evaluating whether Latent Profile Analysis can generate more behaviourally rich, interpretable, and practically useful user groupings for persona creation compared to traditional clustering methods (k-means and hierarchical). By applying these clustering techniques across three datasets and assessing their outputs using a custom evaluation framework, this research aims to provide foundational insight into the potential of model-based clustering in supporting early-stage data-driven persona development in UX/UI and HCI.

Chapter 3

3. Methodology

3.1 Research Approach Overview

This study adopts a comparative evaluation methodology to assess and compare the effectiveness of different clustering techniques for behavioural user segmentation in user persona development. More specifically, this thesis compares two widely used distance-based clustering techniques against a probabilistic model-based clustering method, k-means clustering, agglomerative hierarchical clustering and Latent Profile Analysis (LPA), respectively.

The overall objective of the methodology is to examine whether LPA produces more behaviourally meaningful and interpretable user segments than traditional clustering approaches when applied to UX-related behavioural datasets (Gao et al, 2024). To achieve this, the research was structured into five main phases:

1. Dataset selection,
2. Dataset preprocessing,
3. Implementation of clustering techniques,
4. Cluster evaluation, and
5. Comparative analysis using an Evaluation Framework.

3.2 Dataset Selection Strategy

This initial step involved the selection of suitable behavioural datasets that included relevant, usable user segmentation data (Patkar & Seyff, 2023). Each dataset was selected to represent distinct structures to explore and evaluate the effectiveness of each clustering method against. All the different applications allowed for the assessment of how well the clustering methods generalise across different types of behavioural data.

The following criteria was used to guide the dataset selection, including:

- Relevance to behaviour user segmentation,
- Adequate sample size for clustering analysis,
- Presence of behavioural or interaction variables,
- Compatibility with all clustering methods evaluated, and
- Diversity of data contexts to test the generalisability of results (Salminen et al., 2021; Jansen et al., 2021).

Three datasets were selected, representing simulated behavioural data, real-world behavioural data, and perceptual UX interaction data. These datasets are summarised in *Table 3.1* below.

Table 3.1: Explored Datasets Summary (Faria et al., 2020; Nath, 2025; Rahman et al., 2024) .

Dataset Name	Description	# of Participants	Behavioural Variables	Context	Purpose
Student Habits vs Academic Performance	Simulated study habits, and lifestyle behaviours with academic performance outcomes.	1,000	7	Simulated behavioural data.	Used for baseline testing and to assess whether clustering can uncover behaviourally meaningful simulated data.
Behavioural Classes Brazilian Adolescents	Real collected data from Brazilian adolescents.	217	8	Real-world behaviour data.	Comparison dataset to test the generality of the pipeline against a more coherent, real dataset.
UI/UX User Interaction Dataset Across Popular Digital Platforms	UX UI usability survey and demographics data	2,271	17	UX Perceptual interaction data.	Used to test the effectiveness of LPA on a highly numerical UX dataset.

3.3 Dataset Preprocessing & Behavioural Variable Selection

Secondly, all datasets underwent a variety of preprocessing procedures to ensure comparability and compatibility across clustering methods whilst maintaining meaningful behavioural patterns (Liang et al., 2024; Liu et al., 2022). The preprocessing pipelines included data cleaning and transformation, handling missing or inconsistent values, behavioural feature selection, feature scaling, and dataset-specific transformations such as applying factor analysis.

Considering our final step of producing clusters ideal for user-persona development, the most critical step involved correctly identifying and isolating the proper behavioural variables to ultimately focus on behaviour-driven segmentation rather than demographic groupings such as age or gender.

To ensure comparability across clustering methods, all behavioural variables were standardised using z-score normalisation, allowing each feature to contribute equally during clustering (Quan et al., 2015). For the UI/UX dataset, factor analysis was additionally applied to reduce correlated usability variables into interpretable latent dimensions (Wongabut et al., 2025).

Additional detailed preprocessing steps for each dataset are described in Chapter 4.

3.4 Implementation of Clustering Techniques

Thirdly, the clustering models were implemented using three different clustering approaches: k-means, agglomerative hierarchical clustering, and latent profile analysis. After preprocessing, all clustering methods follow the same general structure between datasets, where each method can be broken down into the general steps detailed below:

- Testing multiple values of cluster number, ' k ',
- Selecting the optimal number of clusters, ' k ', using silhouette scores, elbow plots, and model fit metrics such as BIC & AIC (Spurk et al., 2020; Williams et al., 2016),
- Fitting the final model, and
- Assigning cluster labels.

The following cluster structures were evaluated using quantitative indicators, including cluster balance, behavioural separation, and model fit metrics. The main aim of this stage is to identify the most appropriate number of clusters for each method and dataset.

3.5. Evaluation Framework

A systematic approach was taken to compare clustering methods, detailed in an evaluation framework based on three key dimensions:

- Model performance,
- Behavioural richness, and
- Interpretability.

These dimensions were selected to align with the research aim of assessing the usefulness of clustering methods for persona development.

Model performance focuses on how well clusters are formed and balanced, behavioural richness examines the depth and nuance of behavioural patterns captured, and interpretability evaluates how easily clusters can be understood and translated into meaningful personas (Gao et al., 2023).

Table 3.2: Evaluation Framework.

Dimension	Metric	What it reveals	How it will be used in evaluation
Model Performance	Silhouette Score	Cluster cohesion and separation	To compare performance
Model Performance	BIC	Model fit quality	Select the best-fitting LPA model
Model Performance	Number of clusters	Does LPA produce more/fewer segments?	To compare the segment count across models
Model Performance	Cluster size variance	Are user clusters more balanced or skewed?	To identify dominant or underrepresented groups
Behavioural Richness	Overlap probability	Can LPA capture hybrid users?	To assess probabilistic membership between clusters
Behavioural Richness	Entropy Score	How confident are these user classifications?	To check the certainty of user assignment to clusters
Behavioural Richness	Behavioural Separation	How distinct are the traits across clusters?	To plot or summarise trait differences
Interpretability	Interpretability Assessment	How clear are the raw clusters? Are they suitable for persona creation?	To judge whether clusters can be meaningfully interpreted as UX personas.

3.6.1 Comparative Synthesis of Findings

Lastly, the outcomes of the quantitative cluster analysis will be used to assess whether LPA offers any benefits over traditional clustering techniques for user persona development.

Chapter 4

4. Implementation

4.1 Datasets

To test the research hypothesis, this research project selected 3 datasets across different behavioural contexts. The datasets were chosen to represent varying types of behavioural information and to evaluate whether clustering methods produce consistent segmentation patterns across different data structures.

The three datasets mentioned in Chapter 3 and used in this study are described in more detail below.

4.1.1 Student Habits vs Academic Performance Dataset

The “*Student Habits vs Academic Performance*” dataset contains simulated behavioural data describes student routines, study habits, and lifestyle factors alongside academic outcomes (Nath, 2025). The dataset contains 1,000 participants and includes behavioural variables such as study hours, sleep duration, social media use, class attendance, and physical activity. While demographic attributes are present in the dataset, these variables were excluded from clustering analysis in order to prioritise behaviour-driven segmentation, consistent with best practices in behavioural persona development.

This dataset was included to explore whether clustering techniques could identify meaningful behaviour patterns within simulated lifestyle data and reveal latent student groupings based on study behaviour and daily routines.

4.1.2 Behavioural Classes of Brazilian Adolescents Dataset

The “Behavioural Classes of Brazilian Adolescents” dataset contains real behavioural measurements collected in collaboration between the Ohio University and the Federal University of Viscosa in Brazil (Faria et al., 2020).

The dataset included 217 participants and contained 24 variables, including eight behavioural measures of physical activity, sedentary behaviour, sleep patterns, screen time, and lifestyle risk behaviours such as alcohol and tobacco use. This dataset provides a behaviour-rich context suitable for segmentation analysis and was selected to evaluate whether clustering methods can uncover distinct lifestyle behaviour patterns in real-world health and activity data.

4.1.3 UI/UX User Interaction Dataset

This third and final dataset analysed in this study is the “*UI/UX User Interaction Dataset Across Popular Digital Platforms*” dataset. This dataset contains responses from 2,271 participants who evaluated different aspects of user interface usability across digital platforms (Rahman et al., 2024).

Initially, the dataset included 17 usability rating variables based on Likert-scale responses. These usability ratings examined various aspects of interaction quality, including accessibility, visual hierarchy, navigation, animation, and error handling. Initial clustering attempts using the 17 variables produced difficult-to-interpret clusters due to strong correlations between several variables. To address this issue, factor analysis was applied to reduce the 17 usability measures into nine interpretable latent factors that represent broader usability dimensions.

Due to the application of factor analysis, this dataset differs from the first two datasets as it focuses on and measures interaction quality perceptions rather than behavioural routines. As a result, this dataset provides contrast for evaluating whether clustering methods perform differently across behavioural versus perceptual UX data.

4.2 Preprocessing Implementation

Prior to clustering, each dataset underwent preprocessing to ensure its data were suitable and compatible for segmentation analysis and comparable enough to ensure consistency across clustering methods. As all three datasets originate from different domains and vary in whether they contain real or invented data, each dataset contains variables with different formats, scales, and measurement structures. This made the preprocessing steps crucial to achieving successful final segmentation results.

All scripts followed the same general preprocessing framework, including data cleaning, behavioural feature selection, variable transformation, and feature scaling. Additionally, all datasets were preprocessed uniquely based on dataset-specific requirements and variable metrics to ensure consistency.

4.2.1 Cleaning and transformations

Data cleaning and transformation remained the simplest step in the preprocessing pipeline. It involved removing identifiers, handling missing or invalid values, and converting variables to more consistent formats that would aid the subsequent clustering process. For example, in the Brazilian adolescents dataset, transformations involved converting the screentime and sleep duration variables from hour formats to consistent minute-count formats. Additionally, where initial versions of the alcohol and tobacco use variables were recorded as “yes” or “no,” they were updated to be represented as binary numbers. These minor transformations ensured the behavioural variables could be easily interpreted and processed by the separate k-means, hierarchical and LPA clustering pipelines.

4.2.2 Behavioural feature selection

Within feature selection, only the variables representing behavioural patterns or interaction characteristics were favoured and retained for clustering. Demographic variables, such as gender, age and ethnicity, were excluded from preprocessing to produce clean clustering inputs. This step aimed to ensure the project's scope remained focused on identifying whether behaviour-driven user segmentation performed better than demographic-driven groupings. As seen in *Table 4.1* (refer to appendix), the behavioural variables favoured included lifestyle behaviours such as physical activity, sleep patterns, and study patterns, as well as usability interaction measures that indicate behavioural habits. This section of preprocessing can vary and is somewhat subjective; however, by utilising the same standardised datasets across each clustering method, simple feature selection steps minimise methodological variability and maximise consistency and replicability. For example, the number of behavioural variables selected varied across datasets, depending on the characteristics of each dataset, but all variables were standardised and saved in separate files to prevent changes prior to cluster analysis.

4.2.3 Feature Scaling

Following feature selection, feature scaling was applied to the behavioural variables as clustering within the k-means and hierarchical algorithms is highly sensitive to differences in variable scale. Without scaling, clustering results are likely to be disproportionately influenced and skewed. Z-score normalisation was applied to variables to ensure that features are centred around a zero mean and scaled by individual standard deviation. As a result, the scaling step produced behavioural variables that contributed equally within the clustering process.

4.2.4 Factor Analysis for the UX dataset

An additional preprocessing step was taken for the “*UI/UX Across Popular Digital Platforms*” dataset. This dataset originally included 17 usability rating variables derived from Likert-style response data. Initial clustering attempts using these 17 variables were not easily interpretable due to the high correlations between several usability rating variables. By utilising factor analysis, the variable count was reduced to nine usability factors, yielding better latent categories of interaction that more accurately described and grouped participant behaviour. Factor extraction was performed using a varimax-rotated factor analysis on standardized variables, where variables were grouped into factors based on their highest loading values. For further insight into raw loading variables and their distributions across factors, refer to *Table 4.3* in the appendix.

After detailed analysis of the UX/UI variables that made up the factorised variables, the final 9 interpretable factors are summarised in *Table 4.2*. Each factor is composed of and characterised by the variables with the highest loadings pertaining to it. Consequently, the interpretability and stability of the generated cluster profiles improved cluster richness and complexity, which is pivotal towards the development of good user personas.

Table 4.2: Factorised Behavioural Variables from the “UI/UX User Interaction Across Popular Digital Platforms” dataset.

Factor	Meaning	Based on high loadings
Factor 1	Touch & interaction controls	Gestures, touch controls
Factor 2	Accessibility	Accessibility
Factor 3	Feedback & error clarity	Feedback & error messages
Factor 4	Personalisation	Personalisation
Factor 5	Performance/speed	Loading speed
Factor 6	Functional usability	Forms, search functionality
Factor 7	Motion & transitions	Animations
Factor 8	Social & integration features	Social media integration, hierarchy
Factor 9	Visual design quality	Layout, images, responsiveness

4.2.5 Final prepared dataset outputs

The preprocessing stage produced cleaned, standardised feature matrices for each dataset. The preprocessed variables ensured that each dataset could be analysed consistently while preserving the behavioural characteristics necessary for meaningful segmentation.

4.3 Clustering Implementation

Each clustering method was applied using the same preprocessed datasets. To ensure consistency across methods and datasets, all models were initialised with $n_init=20$ and a $random_state=42$.

4.3.1 K-means

K-means clustering partitions data into clusters by minimising the distance between data points and cluster centroids. Implementation involved:

1. Testing multiple values of k
2. Selecting an optimal k using silhouette scores and elbow plots
3. Fitting the final model
4. Assigning cluster labels

The resulting outputs included the scaled cluster centroids, labelled datasets, PCA visualisations, and profile line plots.

4.3.2. Agglomerative Hierarchical

Agglomerative hierarchical clustering builds clusters using a bottom-up approach. The implementation involved:

1. computing a linkage matrix using Ward's method
2. visualising cluster structure using a dendrogram
3. selecting cluster size using silhouette scores
4. assigning cluster labels

With result outputs including dendrogram visualisations, cluster assignments, PCA plots, and behavioural summaries.

4.3.3 Latent Profile Analysis

LPA was implemented using a Gaussian Mixture Model (GMM) to model latent behavioural profiles. Unlike traditional clustering, LPA assigns probabilistic membership to clusters, thus, allowing it to capture overlapping behavioural patterns. Implementation steps included:

1. testing multiple cluster sizes
2. evaluating models using BIC, AIC, silhouette score, and entropy
3. selecting optimal ' k ' (with manual override where necessary)
4. fitting the final model

The final model assigned each data point a most likely profile (hard label) to better compare results against traditional models. Outputs included latent profile summaries, probability distribution summaries, PCA visualisations and profile line plots.

4.4 Cluster Number Selection

Cluster sizes were consistently selected across segmentation methods and across datasets. Outputs generated with a range of k values (2-10) were used to produce plots to guide final model selection. This range was chosen because 2 is the minimum cluster size that could be realistically analysed, and any cluster size above 10 could likely indicate over-segmentation.

In conjunction with *Table 4.4*, the optimal number of clusters were selected using the following metrics:

- **Elbow method:** to observe for a distinct curve or “elbow” on a plot depicting the Sum of Squared Errors (SSE) that signifies a threshold where adding more clusters no longer improves model fit
- **Silhouette score:** observe for the score closest to 1, which indicates the data points are well-separated
- **Entropy:** observe for high entropy scores (0.60 - 0.80) as it indicates better classification quality,
- **BIC/AIC:** observe for lower values that indicate better model fit, as it evaluates the quality of the model.

Table 4.4: Overview of the cluster size metrics used for each clustering method.

Method	K-means	Hierarchical	LPA
Elbow method	Yes	No	No
Silhouette score	Yes	Yes	Yes
Entropy	No	No	Yes
BIC/AIC	No	No	Yes

In addition to the guidance provided by these methods, the final decision on the optimal number of clusters was subject to change based on the evaluation of the profiles produced, aiming to maximise cluster balance, interpretability, and behavioural richness while minimising ambiguous or small profiles. This step consistently highlighted a trade-off between statistical fit and practical usability, particularly with LPA profiles, as the optimal number of clusters was consistently larger than those indicated by traditional methods. The back-and-forth process for selecting the optimal number of clusters for each dataset is explored in the following sections.

4.4.1 Cluster Size Selection for “*Student Habits vs Academic Performance*”

4.4.1.1 K-means

As seen in *Figure 4.1* below, the downward trend for SSE is consistent and non-sporadic as ‘ k ’ clusters increase. With no distinct slope or “elbow”, this indicates there is no explicit ‘ k ’ point that displays a plateau where increasing the number of clusters improves model fit. Using *Figure 4.1*, the greatest decrease in slope for SSE occurs between clusters $k=2$ and $k=3$; however, this observation is highly subjective, considering this slope is still very light/unnoticeable.

Figure 4.1: K-means SSE Plot for ‘Student Habits vs Academic Performance’ dataset.

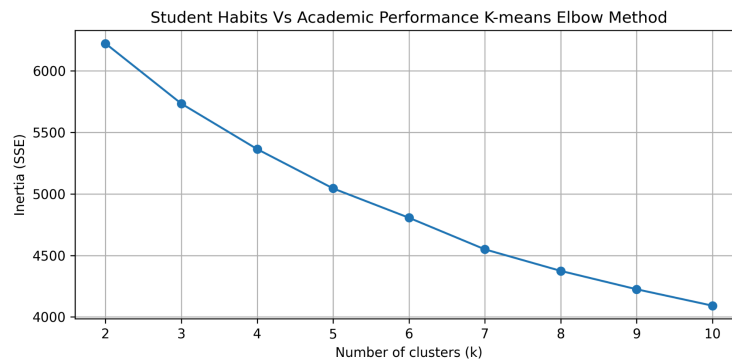
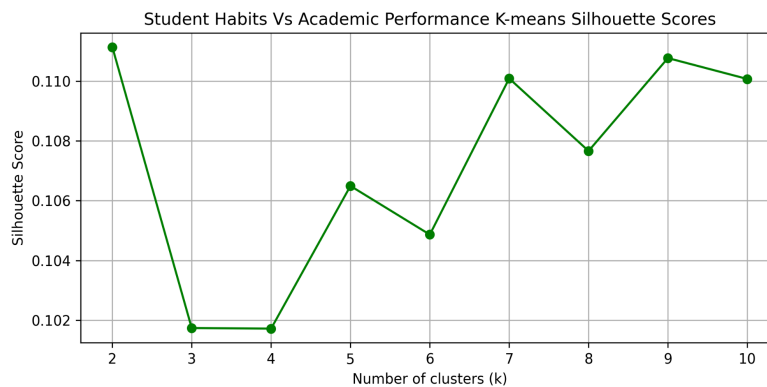


Figure 4.2 shows low Silhouette scores across all cluster sizes, which suggests k-means was not a good clustering method for segmenting behavioural data in this dataset, as scores close to 0 indicate that the model profiles are very close to decision boundaries. Additionally, the short score range (0.102 - 0.110) does not demonstrate a substantial structure to base an optimal cluster number on; however, it does favour values equal to or greater than 5. Because of this, further analysis utilising PCA and Profile Plots was needed to finalise the selection of cluster size, ultimately selecting $k=4$.

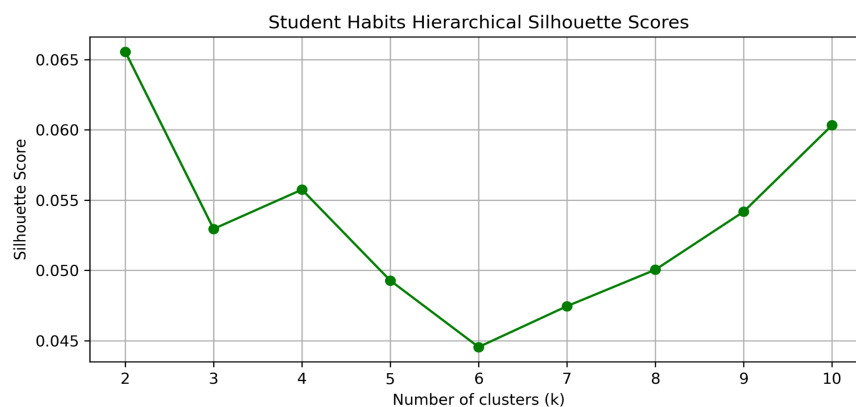
Figure 4.2: K-means Silhouette Plot for ‘Student Habits vs Academic Performance’ dataset.



4.4.1.2 Agglomerative Hierarchical

Similarly to *Figure 4.2*, the silhouette results in *Figure 4.3* show significantly lower scores for the agglomerative hierarchical clustering method, which indicates that across all numbers of clusters ‘ k ’, individual data point separation is minimal and the produced clusters are likely oversegmented and/or not well-matched to their assigned cluster. In order to minimise this effect, values of ‘ k ’ equal to or greater than 5 will not be considered as silhouette scores only become comparable to lower ‘ k ’ results at $k=9$, where resulting small cluster populations can suggest higher oversegmentation likelihood. As $k=2$ is objectively too small to produce insightful profiles usable towards user persona development, the final choice of optimal clusters concludes between $k=3$ and $k=4$. Overall, $k=4$ was selected as the best fit for this clustering method after analysis of the population distributions which showed better cluster balance with cluster sizes ranging between 15.9% and 31.5%. Picking $k=4$ as the optimal number of clusters also confirmed the cluster number of k-means, which was also deduced to be optimal at $k=4$.

Figure 4.3: Hierarchical Silhouette Plot for ‘Student Habits vs Academic Performance Dataset’.

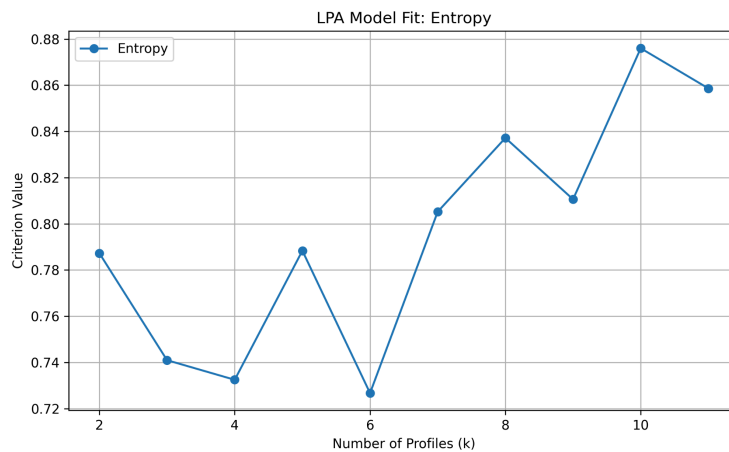


4.4.1.3 Latent Profile Analysis

Figure 4.4 shows that entropy scores below $k=7$ are less stable and have noticeable fluctuations between values. After $k=5$, the entropy scores begin to stabilise and trend upwards, with the exception of $k=6$, which is noted as the last significant downward fluctuation on the plot. Figure 4.4 also demonstrates high entropy values (0.72 and greater) across all ' k ' number of profiles; this indicates good model classification and fit, although higher entropy values closer to 1 are preferred.

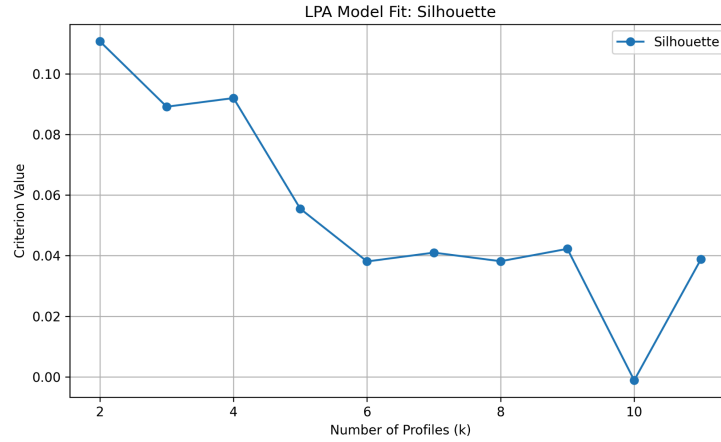
Based on strictly obtaining an optimal ' k ' value that has a corresponding high entropy, the more promising cluster sizes are between $k=5$ and $k=10$. Although the highest entropy score is at $k=10$, such a large cluster size is not ideal and likely indicates over-segmentation. Coupled with existing literature noting entropy scores between 0.60 and 0.80 as the most acceptable, the three promising candidates of ' k ' become $k=5$ (0.79), $k=7$ (0.805) and $k=8$ (0.84).

Figure 4.4: LPA Entropy Plot for 'Student Habits vs Academic Performance Dataset'.



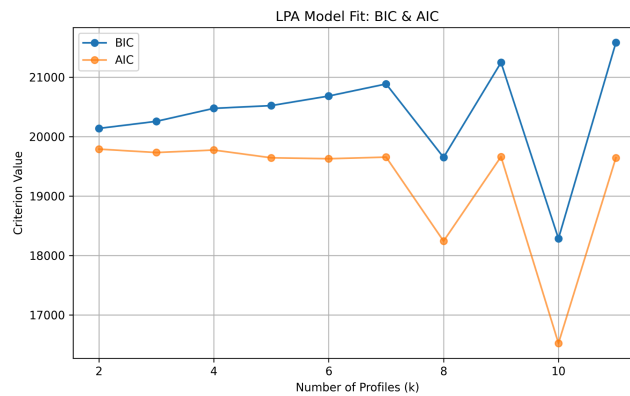
Although the use of silhouette scores is not directly used to pick optimal cluster sizes in LPA, they do help confirm cluster separation based on distance metrics, which is beneficial when comparing probabilistic model cluster results with distance-based partitioning methods such as k-means and hierarchical. Here, Figure 4.5 exhibits silhouette scores decreasing in conjunction with increasing ' k ' number of profiles. There is a noticeable change in slope between $k=4$ and $k=6$, indicating improved cluster separation within this range. Additionally, the plateau beyond $k=6$ suggests that cluster separation stabilises between $k=6$ and $k=9$.

Figure 4.5: LPA Silhouette Plot for ‘Student Habits vs Academic Performance Dataset’.



Lastly, *Figure 4.6* shows the BIC and AIC model fit, and demonstrates relatively stable results between $k=2$ and $k=7$, sequentially followed by noticeable drops at $k=8$ and $k=10$. Whilst explored literature favours low BIC and AIC values, it also notes that BIC is likely to overestimate the number of profiles, particularly in data that's not normally distributed. This raises the possibility that the sharp dips observed at $k=8$ and $k=10$ may not necessarily reflect genuine improved model performance but rather reflect this observed flaw. To reduce the likelihood of this flaw and whilst noting the overall stability provided by smaller cluster numbers, the most reasonable range for cluster selection falls between $k=4$ and $k=8$.

Figure 4.6: LPA BIC & AIC Plot for ‘Student Habits vs Academic Performance Dataset’.



The consensus from all metrics is that the most promising k cluster size is at $k=5$, $k=7$, or $k=8$, as they remained the most consistently performing across findings. Moreover, after analysis of cluster size distribution and potential profile behavioural richness (*see Figure 5.8*), it was revealed that $k=8$ is the most favourable of the options, with only 2 clusters under 10% of the population size and more distinguishable behavioural differences and complexity.

4.4.2 Cluster Size Selection for ‘*Behavioural Classes Brazilian Adolescents*’

Similar trends were observed in the *Behavioural classes for Brazilian Adolescents* dataset. Overall, k-means and hierarchical clustering both suggested $k=3$ based on silhouette scores, while LPA metrics indicated optimal values between $k=4$ and $k=6$. A lower cluster size was preferred to avoid over-segmentation due to the smaller dataset size (217).

4.4.2.1 K-means

Figure 4.7, utilised to perform the elbow method, shows no distinct elbow position when running SSE for k values 2 to 10. There is a very small elbow at $k=3$; however, this is up to individual observation and is a subjective observation.

Additionally, *Figure 4.8* displays this clustering method's silhouette plot, highlighting the highest silhouette score at $k=3$ whilst exhibiting separation between clusters decreases as the ‘ k ’ number of clusters becomes equal to or greater than 4.

4.4.2.2 Agglomerative Hierarchical

Utilising *Figure 4.9*, the most promising silhouette scores occurred at $k=2$ and $k=3$. Subjective analysis led to selecting $k=3$ as the optimal cluster size, as smaller sizes are likely to provide oversimplified results, and larger cluster sizes in this dataset do not provide sufficient enough silhouette scores to justify overly complex results in minimally populated clusters.

4.4.2.3 Latent Profile Analysis

Referring to *Figure 4.10* in the appendix, perfect entropy scores equal to 1 are found for the values between $k=2$ and $k=5$, with a notable drop in slope between $k=5$ and $k=6$, followed by more realistic yet incredibly well-performing entropy scores greater than 0.970, which indicates that all the possible clustering size values showcase high classification quality. However, due to the small dataset population, large cluster sizes are strongly discouraged. Finalising on the value $k=6$ as the optimal cluster number for LPA was due to it being the lowest integer with entropy equal to 1.

Similar to the evaluation of silhouette plot results for the *Student Habits vs Performance* dataset, the silhouette scores for LPA in this dataset do not exhibit significant or beneficial results towards final cluster size selection, as shown in *Figure 4.11* in the appendix.

Lastly, the BIC and AIC plots in *Figure 4.12* indicate that the values between $k=4$ and $k=6$ are the most ideal as cluster sizes, as the model fits the data better in this range than in smaller cluster sizes. *Figure 4.12* also shows an elbow at $k=4$, which indicates that increasing the number of clusters will not significantly improve results.

4.4.3 Cluster Size Selection for ‘UI/UX User Interaction Dataset Across Popular Digital Platforms’

For this dataset, cluster size selection was influenced by the factorised structure of the behavioural variables. After reducing the original 17 UX variables into 9 latent factors during preprocessing, the produced metrics allowed for clustering to be performed on more meaningful, combined behavioural dimensions rather than isolated UX actions. Coupled with this dataset's large size ($n=2,271$), the reduced feature space enhanced the clarity of the structure of the cluster profiles, making cluster patterns more interpretable and easier to evaluate.

Traditional methods showed consistently low silhouette scores that indicated optimal cluster number at $k=4$, while LPA entropy and BIC suggested optimal values between $k=5$ and $k=7$. A moderate cluster size was selected to balance interpretability and behavioural richness.

4.4.3.1 K-means

Similarly to the ‘*Student Habits vs Performance*’ dataset, *Figure 4.13* shows no clear ‘elbow’ point when applying the SSE method, although a very slight objective bend can be observed at $k=4$. Due to the lack of a significant bend, selecting a larger ‘ k ’ number of clusters does not justify an improvement in model fit.

Additionally, the silhouette scores shown in *Figure 4.14* consistently show extremely low scores across all ‘ k ’ values, which does not suggest considerable profile separation through the model. Considering this, silhouette scores notably improve and begin trending upwards past $k=3$.

Based on these observations, $k=4$ was selected as the most reasonable cluster size for k-means, ideally balancing simplicity with limited metric signifying.

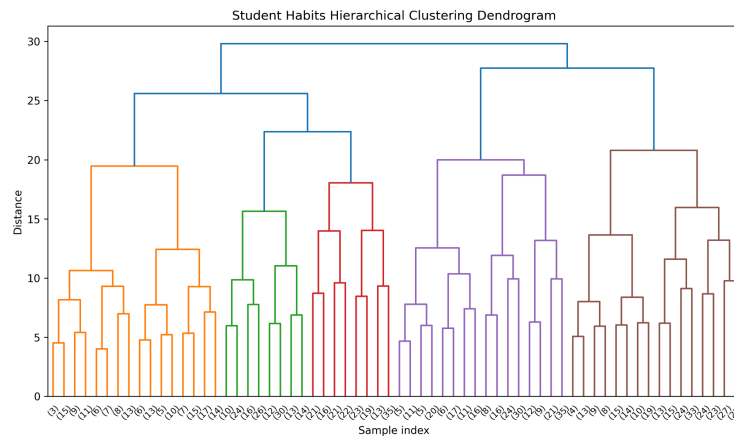
4.4.3.2 Agglomerative Hierarchical

The hierarchical clustering results also follow a similar pattern to the k-means results for this dataset. *Figure 4.15* exhibits consistently low silhouette scores below 0.1, suggesting this application of hierarchical clustering produces poorly separable clusters across all values of ' k '. This suggests that distance-based clustering methods, like k-means and hierarchical, struggle to produce clearly defined clusters when handling factorised behavioural variables. This minimises the complexity offered by these methods for user profile development and introduces a limit for their use with highly numerical and Likert-style data that is central to UX research.

When combining silhouette and dendrogram analysis (see *Figure 4.15* and *Figure 4.16*, respectively), potential cluster structures arise at $k=3$ and $k=5$. However, inspection of cluster balance at $k=5$ reveals uneven cluster sizes with two standout, highly similar smaller clusters (highlighted in green and red within *Figure 4.16*). As a result, $k=4$ was selected as the optimal cluster size for this model, as four larger, balanced clusters are favourable and aim to improve profile richness whilst counteracting low separation.

This decision also aligns with the k-means result, providing consistency across traditional clustering methods.

Figure 4.16: Agglomerative Hierarchical Dendrogram for 'UI/UX Across Popular Digital Platforms'.



4.4.3.3 Latent Profile Analysis

Figure 4.17 displays acceptable entropy results between $k=5$ and $k=10$, with stable entropy around 0.6 between $k=5$ and $k=7$. However, peaks in entropy beyond $k=7$ indicate selecting larger cluster numbers to enhance classification quality, but this improvement is not marginal enough to justify

risking over-segmentation. Additionally, with entropy levels for smaller cluster sizes tipping towards unfavourable, cluster values between $k=5$ and $k=7$ were preferred to balance classification quality, model simplicity and interpretability.

Figure 4.19 shows the lowest BIC and AIC values at $k=2$, with values gradually increasing alongside cluster number. Whilst this may initially suggest that smaller cluster sizes provide a statistically better fit for this model, such significantly low ‘ k ’ values are not suitable for user persona development and would result in overly simplified and uninformative behavioural groupings.

Consistent with previous LPA results across all datasets, *Figure 4.18*’s exhibition of downwards trending, extremely low silhouette scores minimises its influence towards the selection of an optimal cluster number for this dataset.

Given the conflicting nature of these metrics, a balance between model fit, classification quality and interpretability was required. As a result, $k=6$ was selected as the most appropriate cluster size that provided moderate entropy, avoided excessive model complexity and maintained interpretable profiles that still captured meaningful behavioural variation.

4.5 Output Generation and Visualisation

For each clustering method and dataset, a consistent set of outputs was generated to support result analysis. This included labelled datasets, cluster centre summaries that represent average behaviour within clusters, PCA visualisations to aid in analysis of cluster separation and overlap, profile line plots showcasing mean profile behaviour, and structured cluster summaries in JSON format for richer interpretation.

The labelled datasets were created by matching each data point with the corresponding cluster/latent profile. For LPA, this matching was done based on the maximum probability score generated through the Gaussian Mixture Model. The cluster centre summaries were then calculated by averaging each variable within each cluster, providing a representation of the average behavioural profile for each cluster. The cluster centre summaries were then subsequently used to generate profile line plots, which visually highlighted key differences between profiles and respective behavioural characteristics. Additionally, to aid visual analysis of cluster separation and overlap, PCA plots were produced by projecting high-dimensional, labelled data into two dimensions.

Apart from these visualisations, structured JSON files were generated for each cluster to summarise detailed information in tabular form. This includes cluster population counts, demographic distributions, and summary statistics of behavioural variables.

All visualisations and outputs were generated using Python, using libraries such as scikit-learn, pandas, and matplotlib, ensuring consistency across datasets and clustering methods.

Chapter 5

5. Experimental Results

This chapter aims to showcase the basis for evaluating whether LPA produces more behaviourally rich and interpretable user segments for practical persona development compared to traditional clustering techniques, thus addressing the aim of the thesis. In this section, the results across all datasets will be explored and analysed alongside the evaluation framework described in *Table 3.2*.

For each dataset, the produced clustering results of each segmentation method will be accompanied by summary tables outlining the selected ‘*k*’ clustering size, and a further detailed expansion of model performance and behavioural richness through:

- PCA plots to showcase cluster separation, and
- Profile line plots that encapsulate behavioural differences between profiles.

To aid with coherence, *Table 5.0* summarises the final cluster sizes across datasets and highlights key differences in ideal cluster sizes between the segmentation principles, with LPA consistently yielding larger optimal ‘*k*’ values than traditional methods. This presently indicates that LPA cluster profiles provide greater diversity and uncover underlying behavioural patterns that traditional methods did not.

Table 5.0: Final ‘*k*’ cluster size per dataset per method overview.

Dataset	K-means best ‘ <i>k</i> ’	Hierarchical best ‘ <i>k</i> ’	LPA best ‘ <i>k</i> ’
<i>Student Habits vs Academic Performance</i>	4	4	8
<i>Behavioural Classes Brazilian Adolescents</i>	3	3	6
<i>UI/UX User Interaction Dataset Across Popular Digital Platforms</i>	4	4	6

5.1 Student Habits vs Academic Performance Dataset

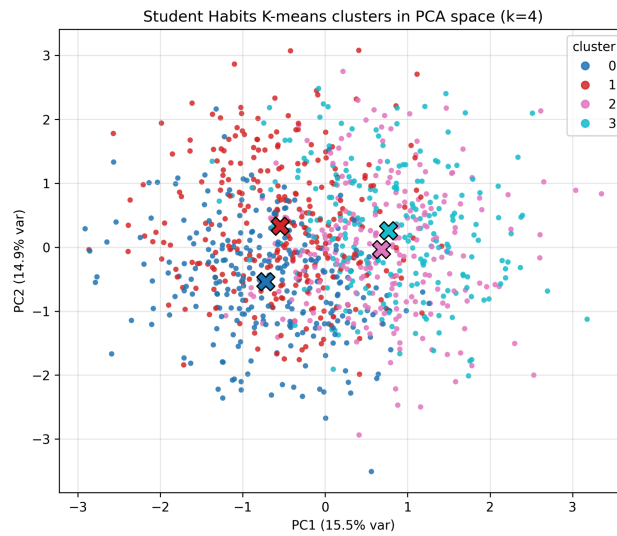
To summarise the clustering results across methods in this dataset, high-performing students were not defined solely by study time but by a combination of significant behaviours such as sleep, exercise, and reduced screen time. Across all methods, LPA produced a greater number of more complex behavioural profiles compared to both traditional methods, however, it introduced an immediate decline in visual interpretability and clarity for the same reason.

Table 5.1: Final ‘k’ cluster size across clustering methods for “Student Habits vs Academic Performance” dataset.

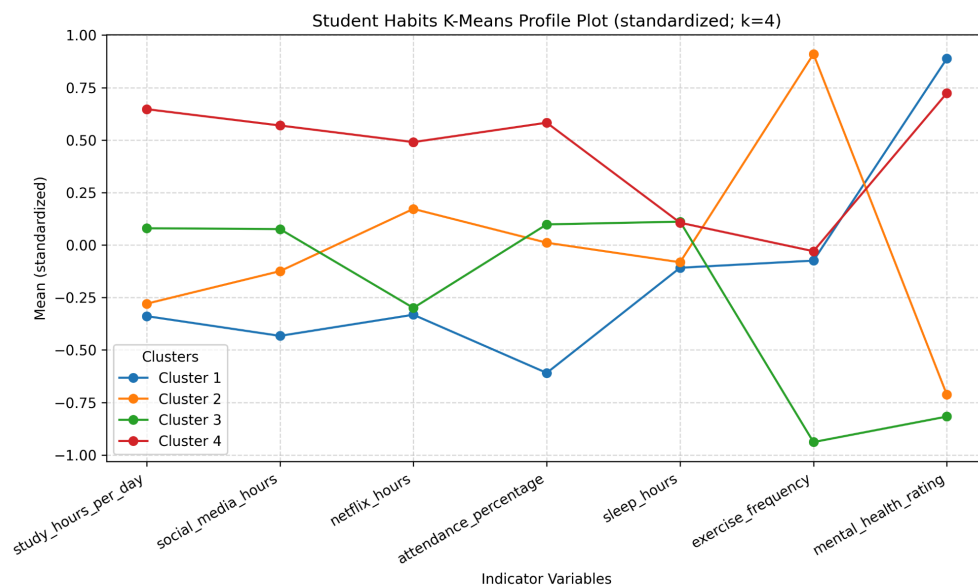
Dataset	K-means best ‘k’	Hierarchical best ‘k’	LPA best ‘k’
<i>Student Habits vs Academic Performance</i>	4	4	8

5.1.1 K-means Results

The k-means clustering results at $k=4$ produced well-balanced clusters, with each profile encompassing approximately 22-28% of the dataset population. This indicates strong model performance that produces stable and reliable segmentation results with minimal dominant or underrepresented profile groups, as shown in *Figure 5.2*. However, the PCA plot in *Figure 5.1* reveals a noticeable overlap between clusters in a reduced feature space, indicating limited separation at the data-point level. This highlights a key limitation of k-means, where simple and interpretable clusters come at the cost of fully capturing nuanced behavioural patterns.

Figure 5.1: K-means PCA plot at $k=4$ for “Student Habits vs Academic Performance” dataset.

The behavioural differences between clusters, shown in *Figure 5.2*, demonstrate clear behavioural distinctions based on polarising values in exercise frequency and mental health, with secondary separation driven by study time and attendance. This is visible in *Figure 5.2*, where the well-performing Cluster 4, characterised by high scores in study hours, attendance, and mental well-being, is directly contrasted with the lower-performing clusters (clusters 2 and 3), with antipodal exercise preferences and low mental well-being scores. Refer to *Figure 5.3* for academic outcomes.

Figure 5.2: K-means Profile line plot at $k=4$ for “Student Habits vs Academic Performance” dataset.

5.1.2 Agglomerative Hierarchical Results

The agglomerative hierarchical clustering at $k=4$ produces moderately balanced clusters, with profile sizes ranging from 15.9-31.5% of the dataset population. Compared to k-means, this model's distributions show a slightly larger imbalance, as evidenced by Cluster 2 housing the largest population, and with slightly more overlap between data points in feature space, as shown in the PCA plot in *Figure 5.4*. This reinforces that while hierarchical clustering does identify meaningful average differences between cluster groups, partitioning between individual data points is not strong enough to showcase substantial behavioural profiles that can be separated into performance groups. This is further reinforced by *Figure 5.6*, which shows minimal differences in the exam score distribution.

The produced clusters, shown in *Figure 5.5*, exhibit clear, easy-to-distinguish behavioural groupings based on variables like social media usage, sleep hours per night and exercise frequency. This is evident through Cluster 3, which shows a grouping of students with the highest social media hours and exercise frequency, and is directly contrasted by Clusters 1 and 2, which display student groupings with opposite scoring in social media engagement and exercise frequency, respectively. Additionally, compared to the clustering produced by k-means, we see a stark contrast in how the sleep variable plays out in segmentation, where previously near-identical sleep points between clusters in *Figure 5.2* are contrasted with the differing results between clusters in *Figure 5.5*.

Overall, agglomerative hierarchical clustering produced moderately interpretable clusters with mostly balanced behavioural variation.

Figure 5.4: Agglomerative Hierarchical PCA plot at $k=4$ for “Student Habits vs Academic Performance” dataset.

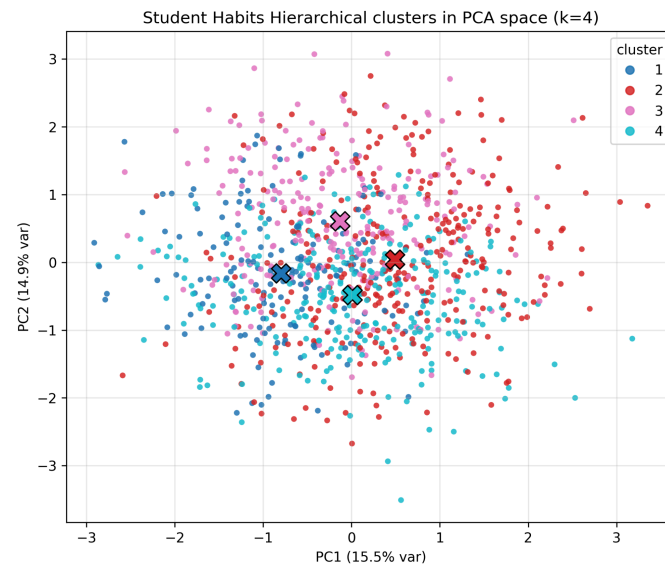
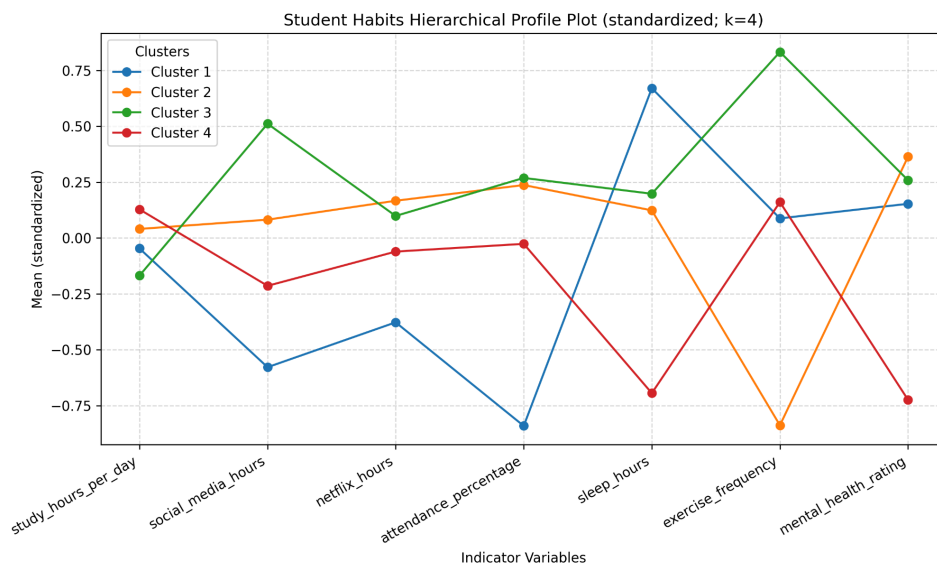


Figure 5.5: Agglomerative Hierarchical Profile line plot at $k=4$ for “Student Habits vs Academic Performance” dataset.



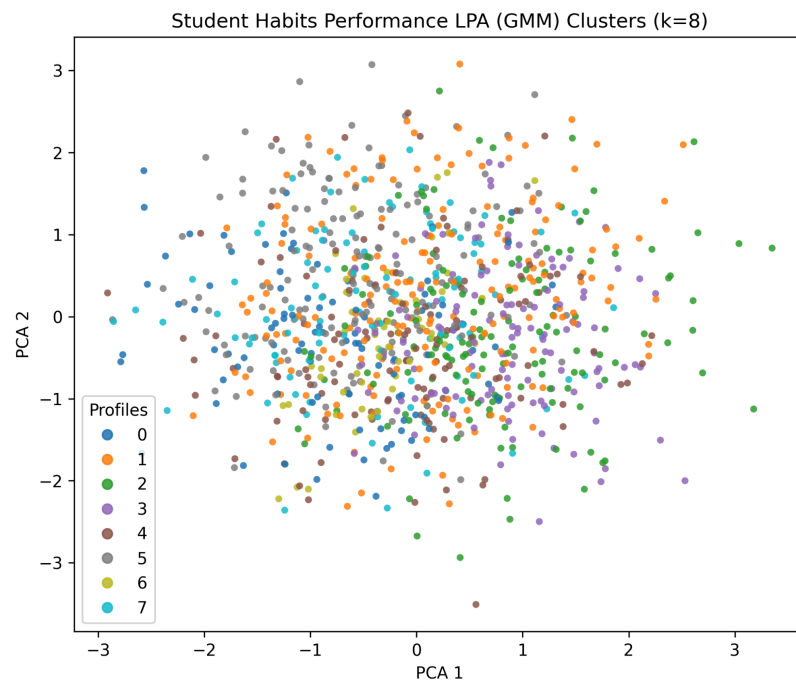
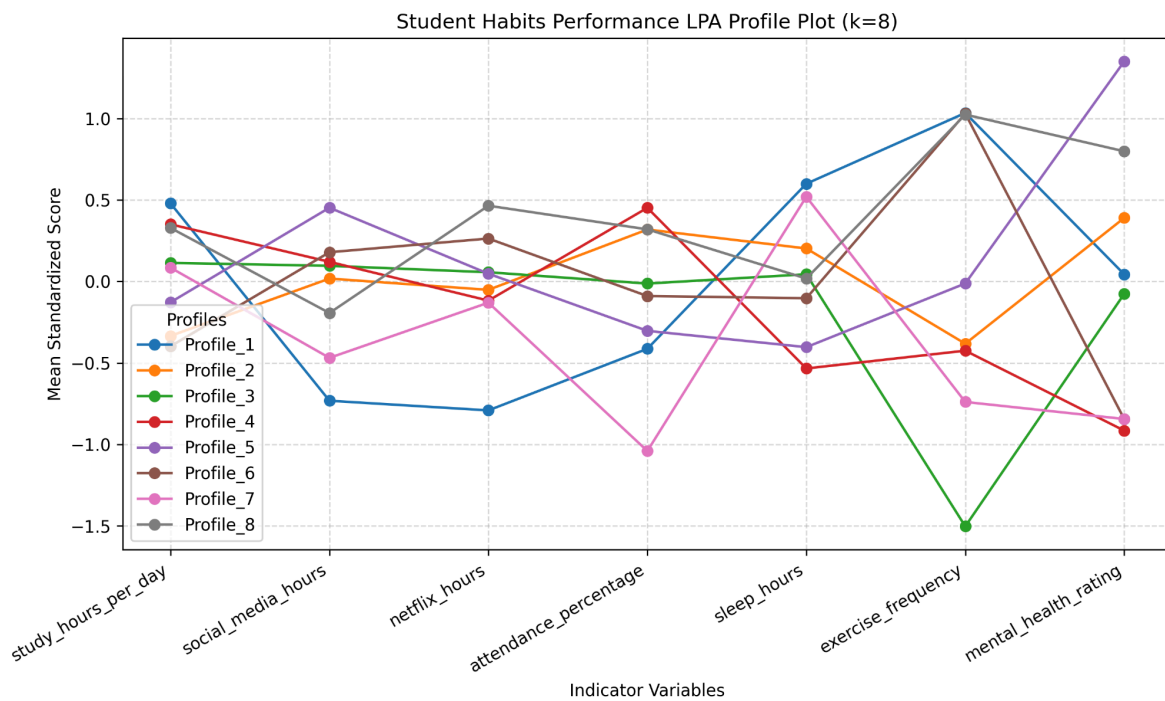
5.1.3 LPA Results

The LPA clustering model at $k=8$ produces more granular segmentation than both traditional methods, k-means and hierarchical clustering, with LPA's cluster sizes ranging from 5.5-18.6%. This distribution is less balanced, indicating several smaller niche profiles.

The LPA profile plot shown in *Figure 5.8* highlights this increased complexity and overlapping behavioural patterns across profiles. Unlike traditional methods, these cluster profiles are not defined solely by extreme behaviour variables and suggest that LPA captures more subtle variations in behaviour that could be useful for user persona creation. A clear example of this can be seen through Clusters 1 and 8, showcasing two profiles of high-performing student groupings whose only similarity encompasses study and exercise habits, whilst providing a niche insight into a well-balanced group of well-performing students with high social media engagement and high mental health ratings which directly contrasts the mental health ratings of Cluster 1, who happen to be the best performing between the two. However, this complexity in behaviour richness creates a trade-off that makes the cluster profiles produced by LPA less easily interpretable.

Additionally, the PCA plot shown in *Figure 5.8* reinforces the substantial overlap between profiles and the weak separation consistent across clustering methods in this dataset.

Overall, LPA produces richer and more detailed behavioural profiles but at the cost of cluster balance and interpretability.

Figure 5.7: LPA PCA plot at $k=8$ for “Student Habits vs Academic Performance” dataset.Figure 5.8: LPA Profile line plot at $k=8$ for “Student Habits vs Academic Performance” dataset.

5.2 Behavioural Classes Brazilian Adolescents Dataset

The clustering trends for the “*Behavioural Classes of Brazilian Adolescents*” dataset were highly comparable to those of the “*Student Habits vs Performance*” dataset. Although cluster distribution imbalance was more apparent due to the smaller sample size ($n = 217$), LPA continued to produce more profiles, each with greater levels of behavioural richness and strong classification certainties when compared to k-means and hierarchical clustering.

Table 5.2: Final ‘k’ cluster size across clustering methods for “Behavioural Classes Brazilian Adolescents” dataset.

Clustering Method	K-means	Agglomerative Hierarchical	LPA
Best ‘k’	3	3	6

Traditional clustering algorithms, such as k-means and hierarchical, yielded easily interpretable and well-separated profiles at $k=3$. These profiles were primarily defined by clear differences in activity levels (indicated by LPA/day, MVPA/day, and steps/day), sedentary behaviour (including screentime and sleep), and lifestyle behaviours (such as alcohol and tobacco consumption). However, this separation followed simple low-medium-high tiers that support limited behavioural richness despite high interpretability. Refer to *Figure 5.11* and *Figure 5.13* for profile line plots.

In contrast, as shown in *Figure 5.15*, LPA identified finer, more granular subgroups with slightly more complex behavioural combinations not identified by traditional methods. However, some clusters were significantly small and unstable, with at least 1 profile containing less than 5% of the population at both $k=4$ and $k=6$, hence, indicating reduced stability and an increased risk of over-clustering. The small size of this dataset does increase the likelihood of minor variations resulting in disproportionately small clusters, however, considering extremely small clusters do not occur at k-means or hierarchical applications at similar cluster sizes like $k=3$, it suggests this was a presentation of LPA’s ability to identify niche and underrepresented subgroups. Coupled with high entropy values greater than 0.975 at $k=4$ and $k=6$, see *Figure 4.10*, this signifies that the efficiency of LPA does not diminish as the dataset shrinks, and consistently generates profiles with strong classification certainty.

Overall, these results suggest that traditional clustering methods are more effective for producing stable and interpretable groupings in smaller datasets, while LPA provides greater behavioural richness at the cost of cluster balance and interpretability. This highlights an important trade-off between simplicity and depth when applying clustering techniques to smaller behavioural datasets.

5.3 UI/UX User Interaction Across Popular Digital Platforms Dataset

The clustering results for this dataset considerably differ from previous datasets due to the factorised structure of the behavioural variables. By reducing 17 UX interaction metrics into 9 latent factors, clustering was performed on more meaningful behavioural dimensions, which allowed for clearer, more interpretable and balanced profile structures. *Table 5.3* shows that both traditional clustering methods, k-means and hierarchical, performed optimally at $k=4$, whilst LPA produced more behavioural insight at $k=6$.

Table 5.3: Final ‘k’ cluster size across clustering methods for “UI/UX User Interaction Across Popular Digital Platforms” dataset.

Clustering Method	K-means	Agglomerative Hierarchical	LPA
Best ‘k’	4	4	6

5.3.1 Traditional Clustering Results (K-means, Agglomerative Hierarchical)

Both k-means and hierarchical clustering produced well-balanced clusters with clear and interpretable behavioural distinctions, primarily separated along dominant UX factors such as accessibility, feedback and error clarity, and performance. PCA visualisations for k-means and hierarchical clusters, shown in *Figure 5.16* and *Figure 5.18*, respectively, demonstrate moderate separation between clusters, with overlap that establishes limited distinction in the feature space.

Despite the similarities between these methods, it is important to note that k-means produced simple profiles that were often largely defined by a single dominant factor, as shown in *Figure 5.17*, thus resulting in clusters that reflect unilateral UX preferences rather than nuanced behavioural patterns. In contrast, hierarchical clustering yielded profiles with more blended and dynamic combinations of UX characteristics, which is preferable for more complex behavioural analysis towards user persona development.

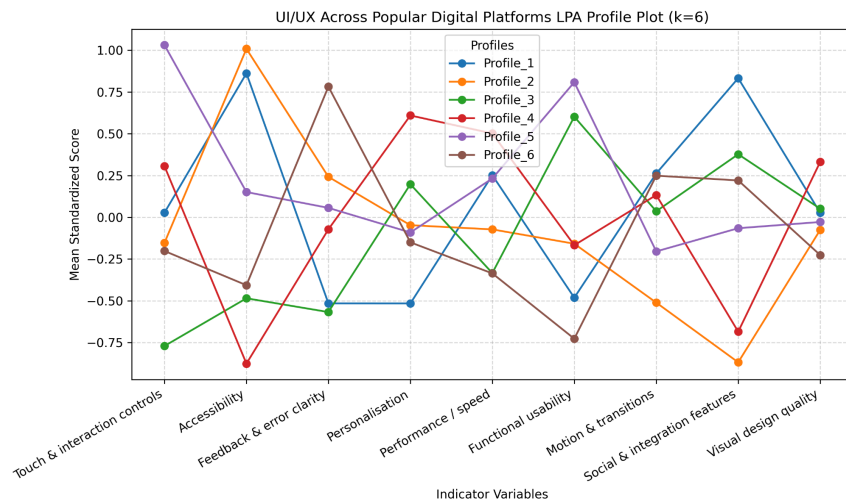
Compared with previous results on other datasets, the application of the factorised UX variables between k-means and hierarchical clustering produced more favourable profile results that were not only relatively easy and quick to interpret and differentiate but also showed more consistent, balanced cluster sizes with varying levels of behavioural richness, ranging from limited to moderate. Overall, this indicates positive initial results for applying factor analysis to highly dimensional numerical Likert UX data in user persona development.

5.3.3 Latent Profile Analysis Results

As expected, LPA produced more behaviourally rich and nuanced profiles, which, unlike traditional methods, separated users based on dominant factors. Complex combinations of UX behaviours characterised LPA clusters across multiple dimensions. LPA's ability to consistently capture more profiles and hybrid user behavioural patterns positively reflects on its potential towards user persona development with real-world UX interaction data.

From a model performance perspective, the cluster size distribution remained very well balanced compared to previous datasets, with all clusters containing 14-20% of the sample, indicating that the larger dataset size helped stabilise LPA outputs despite the increased number of clusters. This was similarly supported by results from k-means and hierarchical clustering, each containing approximately 25% of the sample. Additionally, the moderate entropy of 0.58 at $k=6$ shows LPA outperforms traditional methods by identifying more diverse and overlapping behavioural patterns that maintain acceptable classification confidence. *Figure 5.21* shows LPA profile results at $k=6$, which exhibit greater variation and less uniformity across multiple UX dimensions, supporting the identification of hybrid user types.

Figure 5.21: LPA Profile Line plot at $k=6$ for 'UI/UX User Interaction Across Popular Digital Platforms' dataset.



Regarding interpretability, the factorised variables significantly improved the usability of LPA profiles and their translation into user personas. Prior to factorisation, clustering the original 17 UX/UI variables resulted in highly fragmented, difficult-to-interpret profiles, with individual variables

dominating and failing to form meaningful behavioural groupings, as shown in *Figure 5.22*. By reducing these to 9 latent factors, the clusters instead reflected combined UX behaviours that make profiles more coherent and easier to analyse. For example, *Figure 5.21* shows Profile 5, which depicts users who favour highly interactive and usable interfaces with reliable control and search features. However, despite the model's high positioning in behavioural interpretability, the dataset's limited demographic variables, restricted to age and gender, create an obstacle towards traditional persona development, which relies heavily on this type of contextual data.

It can be concluded that LPA performs well on datasets where behavioural variables are structured into latent dimensions, as it can leverage these relationships to produce both richer and more interpretable user groupings. Thus, making LPA favourable when developing behaviour-based user personas from latent user groupings with minimal demographic data; however, additional demographic or contextual data may be required to fully support traditional persona development in HCI.

5.4 Summary of Findings

Across all datasets explored, traditional clustering techniques yielded better-balanced, easier-to-interpret clusters than those produced by LPA, thereby reinforcing its positioning, suitability, and reliability in persona development.

In contrast, LPA generated more behaviourally rich and nuanced profiles that captured overlapping and hybrid behaviours which could not be easily and justifiably captured using traditional clustering techniques. However, this was sometimes achieved at the expense of reduced cluster balance and interpretability, as observed in multiple instances, and highlighted LPA's sensitivity to dataset characteristics, such as sample size.

The UX dataset results highlight and establish that when behavioural variables are properly structured through factor analysis into latent dimensions, LPA becomes significantly more effective, producing both interpretable and behaviourally meaningful clusters. This cements the key role that preprocessing plays in the success of model-based clustering. However, as clustering across all datasets deliberately excluded demographic attributes, the resulting behavioural groupings reveal limited or overgeneralised demographic patterns that hinder direct and quick translation into traditional personas. This suggests that LPA may be more favourable in development contexts where users are intended to be defined solely by interaction patterns rather than demographic characteristics, or where demographic data is limited or unreliable.

Subsequently, a key trade-off between simplicity and behavioural richness is evident and will be explored further in the following discussion.

Chapter 6

6. Discussion

This study's findings largely align with existing literature on clustering methods for user persona development. Existing research has consistently found that traditional, hard-clustering techniques, such as k-means and agglomerative hierarchical clustering, are effective at producing clear, interpretable groupings (Gao et al., 2023). However, it also notes that these methods cannot capture more complex behaviour-based relationships that depict overlapping traits and uncertainty. This is supported across all datasets, where traditional methods consistently produced more balanced and easily interpretable clusters, primarily separated by a single or very few dominant variables rather than nuanced behavioural patterns. These results reflect limitations previously identified in persona development, where oversimplified groupings can fail to capture the complexity of real users (Hudson, 2013; Siika-aho, 2023).

Additionally, the results supported current literature on Latent Profile Analysis, a model-based clustering method, whose probability-based foundation highlights its ability to identify latent behavioural groupings (Spurk et al., 2020). This analysis was confirmed across datasets, where LPA consistently produced more clusters and identified hybrid behavioural patterns not observed with traditional methods. This aligns with findings in psychology and strongly positions LPA as a suitable approach for application in HCI for user persona development, where profile segmentation requires behaviourally rich user groupings, and robust groupings of vague variables are present or where demographic metrics are limited or unavailable for analysis. As a result, LPA introduces a trade-off in practical usefulness, where a balance between behaviourally rich and interpretable profiles influences how effectively clustering results can be translated into descriptive, actionable user personas.

Overall, the findings suggest LPA is favourable for identifying behaviourally rich user profiles that can be converted into behaviourally supported user personas. Additionally, suggest a potential shift in persona development towards behaviour-based or prioritised segmentation, as underlying behavioural patterns may offer more reliable generalisations on populations rather than self-reported demographic attributes.

Chapter 7

7. Conclusion and Future Work

7.2 Conclusion

This report assesses whether Latent Profile Analysis (LPA) produces more interpretable, behaviourally rich, and practically useful user groupings for persona development than k-means and hierarchical clustering. Analysis of three separate databases using three clustering methods has been recorded, including details of the system pipeline, ranging from preprocessing steps to the best cluster outputs, and evaluation methods and metrics. Limitations and implications of the profile results from Latent Profile Analysis were also identified and justified, including the need for additional analytical methods and the inclusion of demographic data. In conclusion, whilst traditional methods more consistently produce clear, balanced, and interpretable clusters, LPA generates richer, more nuanced profiles that support a shift towards behaviour-driven segmentation in the UX development process. Thereby improving the depth and complexity of user personas whilst providing an avenue for high-level user persona development that requires only analysis of user interaction data, with minimal reliance on demographics and qualitative data retrieval.

7.3 Future Work

Following on from this project, these are some potential research avenues to further explore the practicality of LPA applications in UX persona development:

1. **Hybrid clustering approaches:** Integration of Latent Profile Analysis with a distance-based clustering method, to contextualise the distribution of demographic data within produced LPA behavioural profiles. Ideal results will establish reliable methods that improve interpretability and usability, and simplify the translation of raw clusters into effective, well-rounded user personas that balance behavioural richness with demographic alignment.
2. **Application of factorisation techniques to more diverse databases and contexts:** Further investigation of LPA's effectiveness across a wider range of datasets, particularly those with complex data types and structures, that allow a primary focus on the application of

factorisation into latent variables. Would include exploration of factorisation techniques in a non-UX context, as well as datasets that mix numerical, categorical, and non-Likert variables to capture LPA's performance range and best applications.

3. **Qualitative validation of clustering outputs:** Conducting qualitative validation studies with UX professionals to evaluate the effectiveness and interpretability of raw LPA-generated profile clusters at showcasing insightful behavioural information and their practical translation into real-world, usable user personas. This research avenue would ideally conclude whether LPA's behaviour-driven clusters support design decision-making in applied UX workflows.

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Appendix

Table 4.1: Behavioural features picked with behavioural variables ticked

Student Habits vs Academic Performance	Behavioural Classes of Brazilian Adolescents	UI/UX User Interaction Across Popular Digital Platforms
Study_hours_per_day	Light physical activity per day (min)	Color Scheme
Social_media_hours	Moderate-to-vigorous physical activity per day (min)	Visual Hierarchy
Netflix_hours	Steps/day	Typography
Attendance_percentage	Sedentary Behaviour per day (min)	Images and Multimedia
Sleep_hours	Screen Time	Layout
Exercise_frequency	Sleep duration (hour: min)	Mobile Responsiveness
Mental_health_rating	Alcohol use (1 - no; 2 - yes)	CTA (Call to Action) Buttons
	Tobacco (1 - no; 2 - yes)	Forms and Input Fields
		Feedback and Error Messages
		Loading Speed,
		Personalization
		Accessibility
		Animation and Transitions
		Scrolling_Behavior
		Gestures and Touch Controls
		Search Functionality
		Social_Media_Integration

Table 4.3: Table detailing rounded factor loadings.

UX/UI variable	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6	Factor 7	Factor 8	Factor 9
<i>Colour scheme</i>							0.1		0.12
<i>Visual Hierarchy</i>								0.2	
<i>Typography</i>									
<i>Images and Multimedia</i>									0.14
<i>Layout</i>	0.04					0.1			0.14
<i>Mobile Responsiveness</i>		0.028						0.1	0.19
<i>CTA Buttons</i>									0.07
<i>Forms and Input Fields</i>						0.21	0.1		
<i>Feedback and Error Messages</i>			0.99						
<i>Loading Speed</i>					0.5				0.04
<i>Personalization</i>				0.95					
<i>Accessibility</i>		0.99							
<i>Animation and Transitions</i>							0.35	0.1	0.04
<i>Scrolling Behaviours</i>					0.1		0.11		
<i>Gestures and Touch Controls</i>	0.99								
<i>Search Functionality</i>						0.34			
<i>Social Media Integration</i>								0.23	

Figure 4.7: K-means Elbow Plot for ‘Behavioural Classes Brazilian Adolescents’.

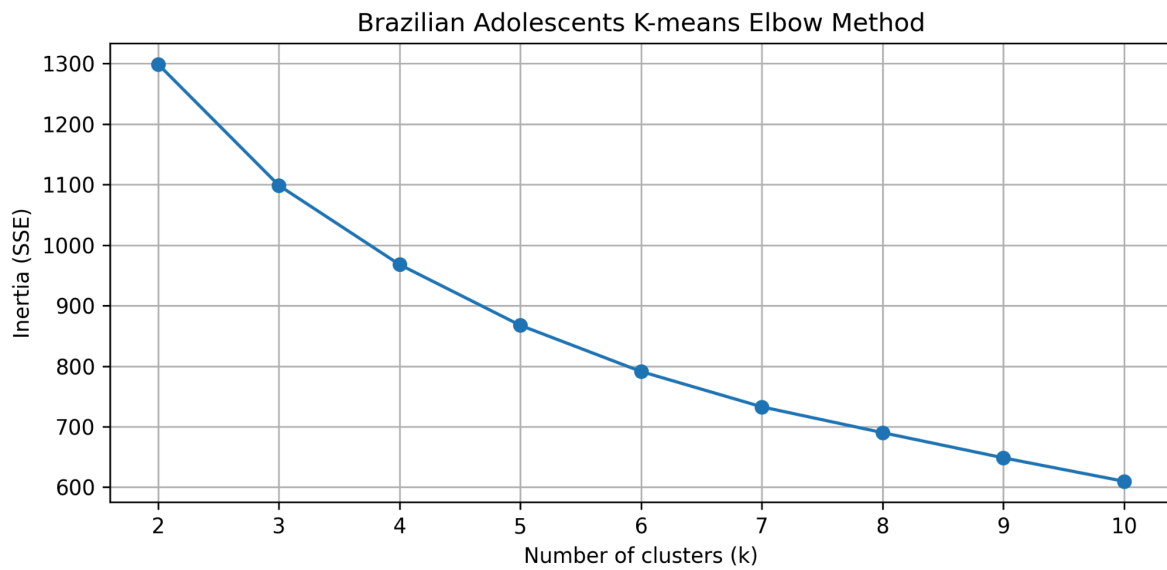


Figure 4.8: K-means Silhouette Plot for ‘Behavioural Classes Brazilian Adolescents’.

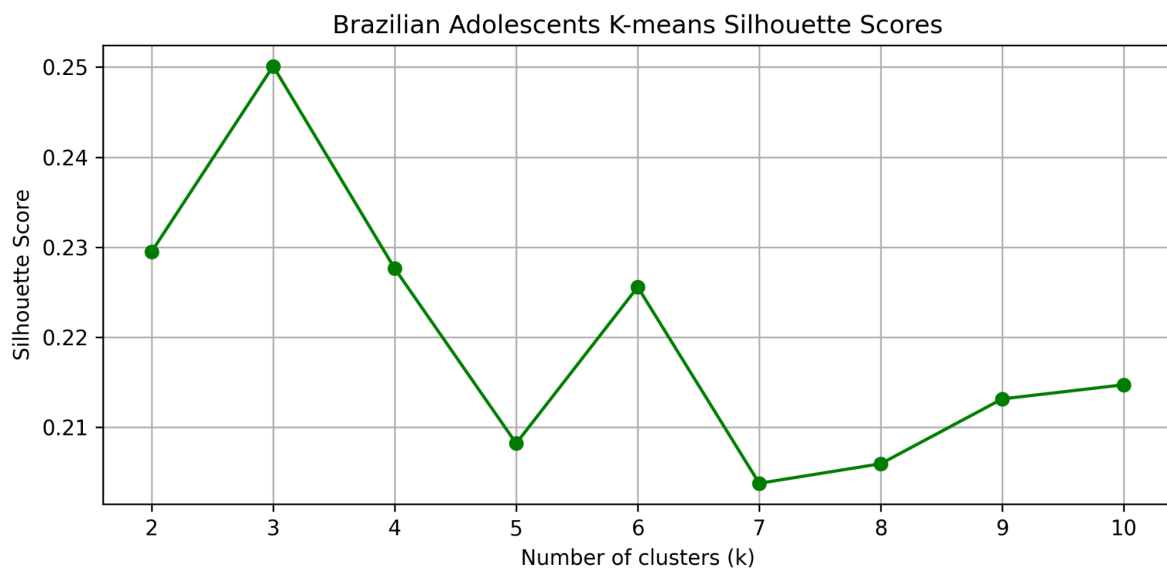


Figure 4.9: Agglomerative Hierarchical Silhouette Plot for ‘Behavioural Classes Brazilian Adolescents’.

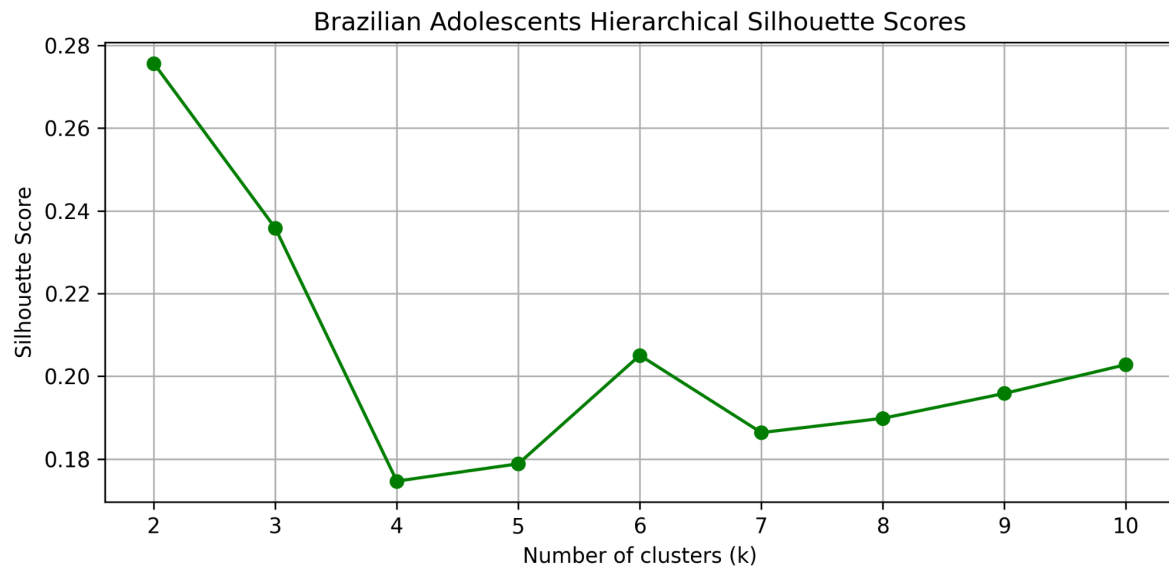


Figure 4.10: LPA Entropy Plot for ‘Behavioural Classes Brazilian Adolescents’.

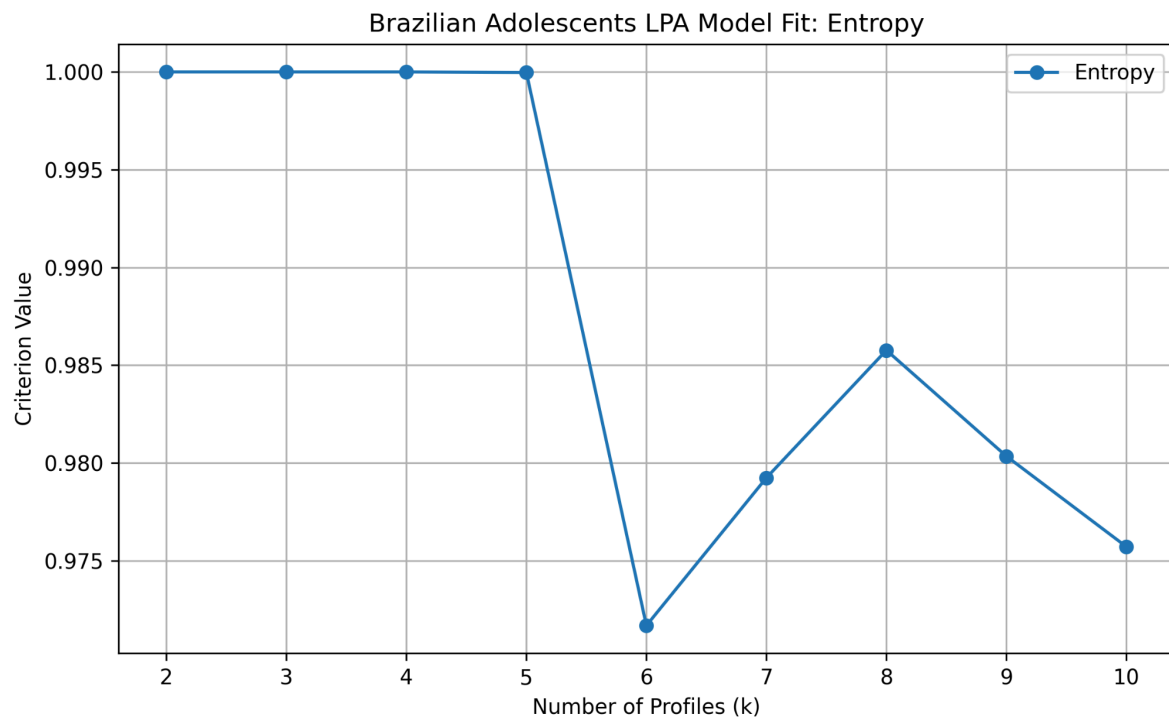


Figure 4.11: LPA Silhouette Plot for ‘Behavioural Classes Brazilian Adolescents’.

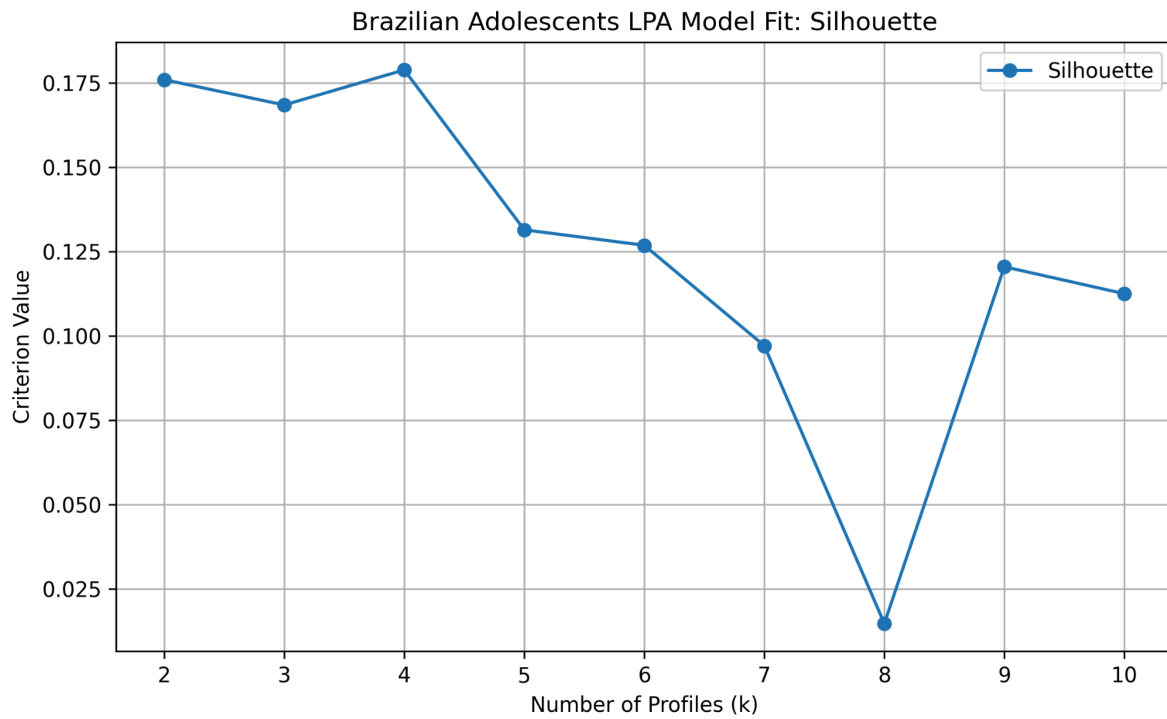


Figure 4.12: LPA BIC & AIC Plot for ‘Behavioural Classes Brazilian Adolescents’.

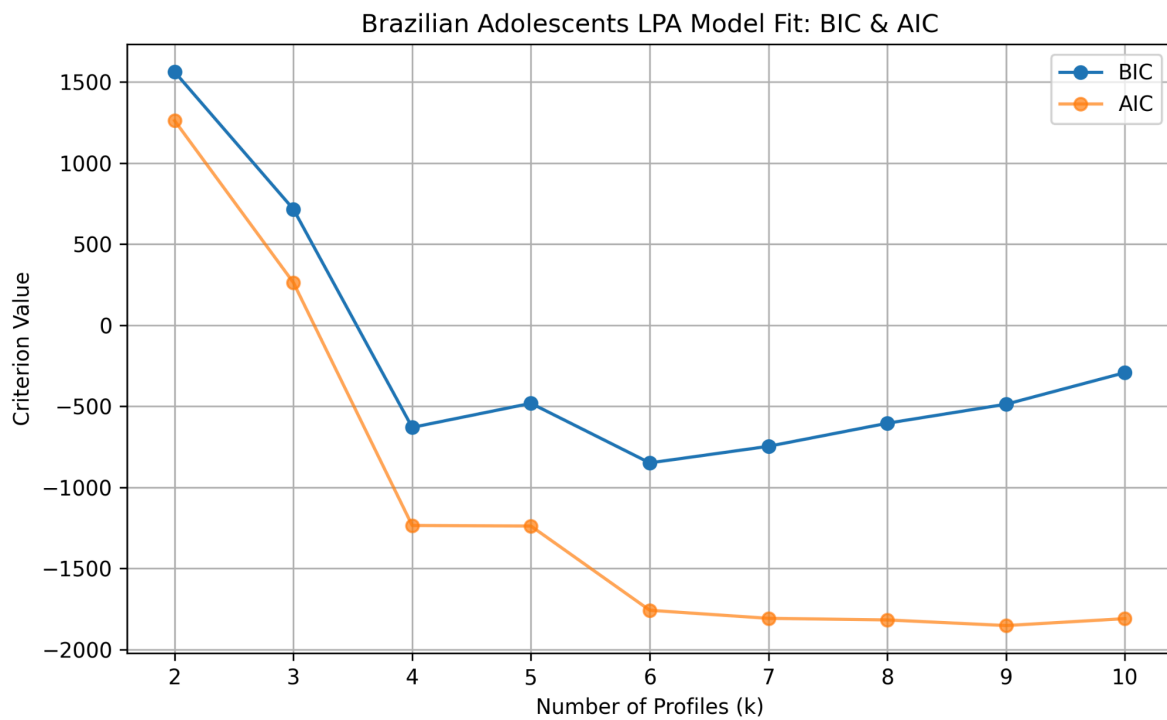


Figure 4.13: K-means Elbow Plot for 'UI/UX Across Popular Digital Platforms'.

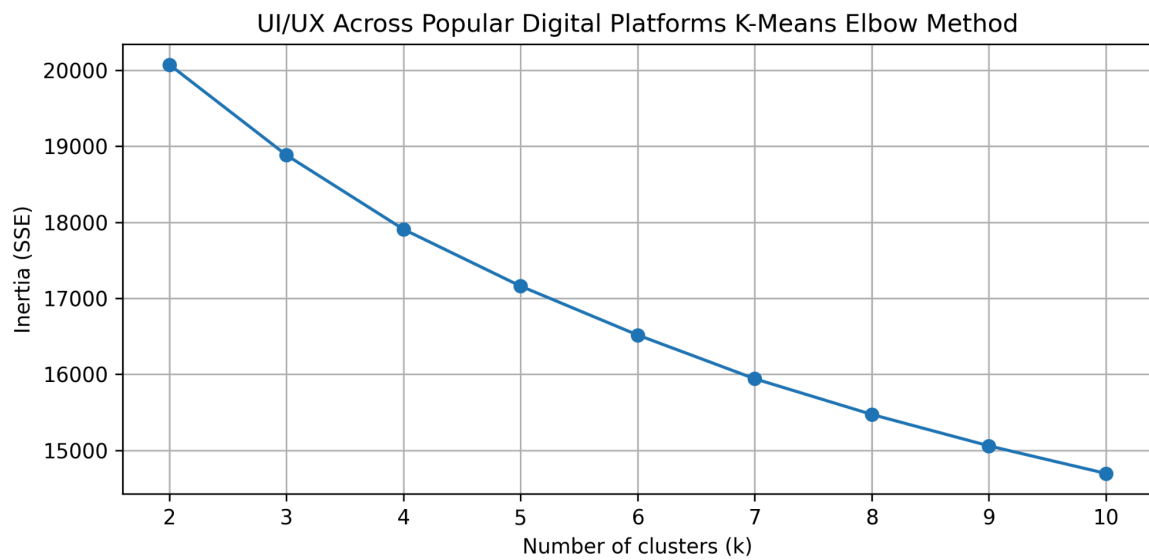


Figure 4.14: K-means Silhouette Plot for 'UI/UX Across Popular Digital Platforms'.

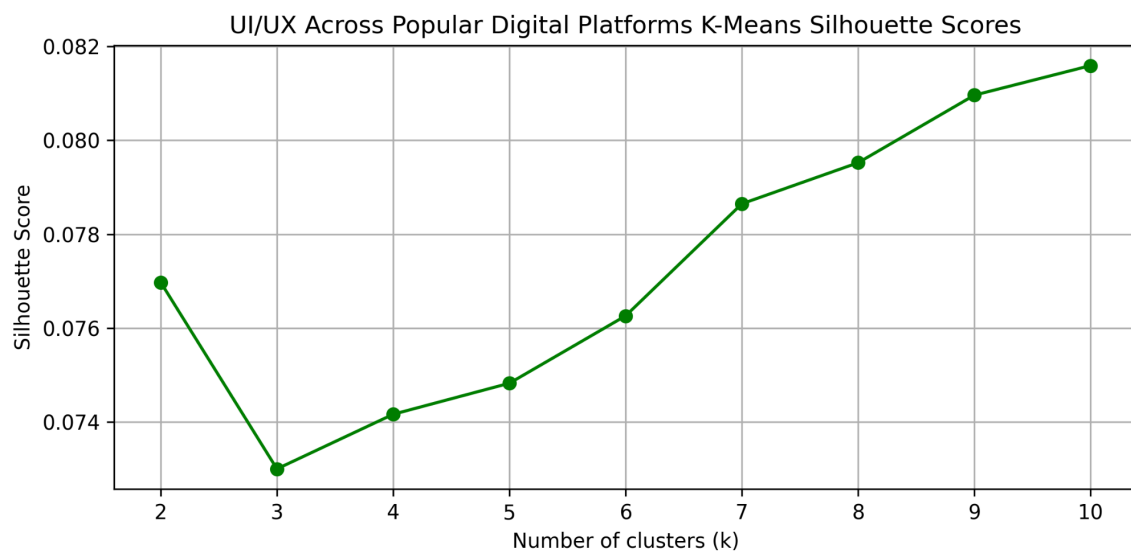


Figure 4.15: Agglomerative Hierarchical Silhouette Plot for ‘UI/UX Across Popular Digital Platforms’.

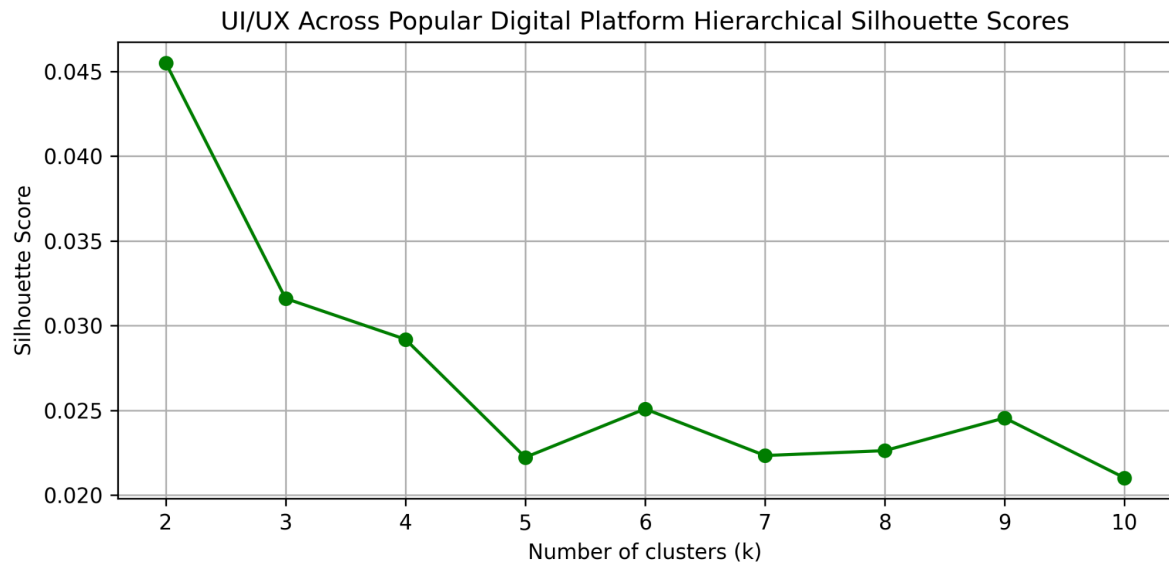


Figure 4.17: LPA Entropy Plot for ‘UI/UX Across Popular Digital Platforms’.

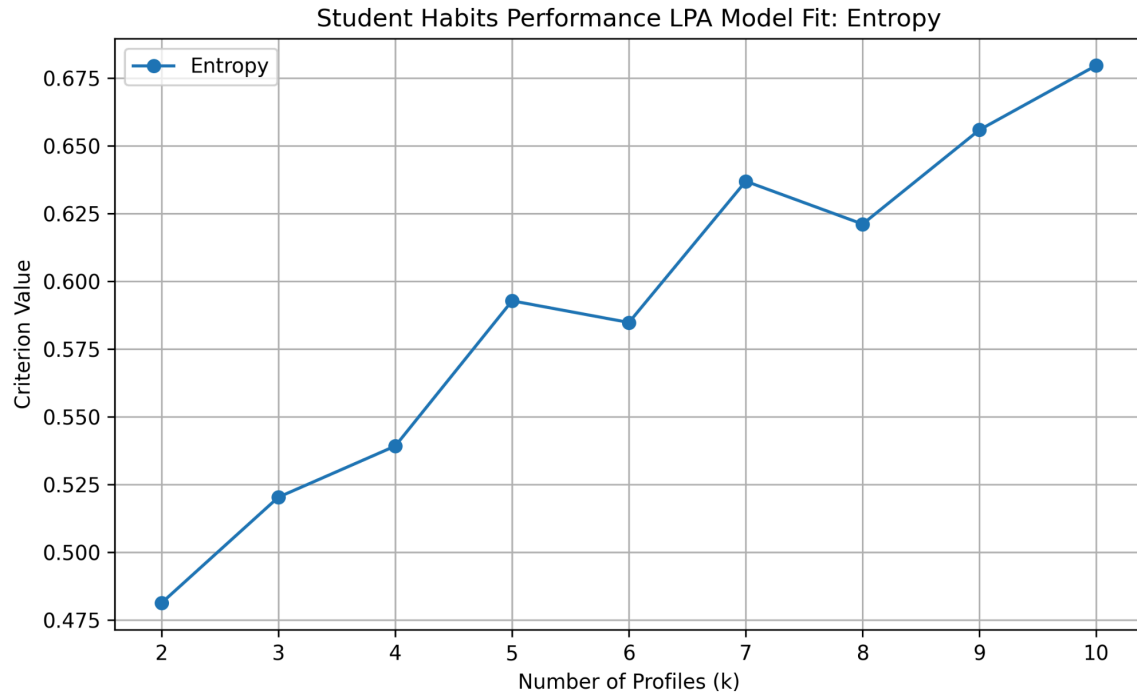


Figure 4.18: LPA Silhouette Plot for 'UI/UX Across Popular Digital Platforms'.

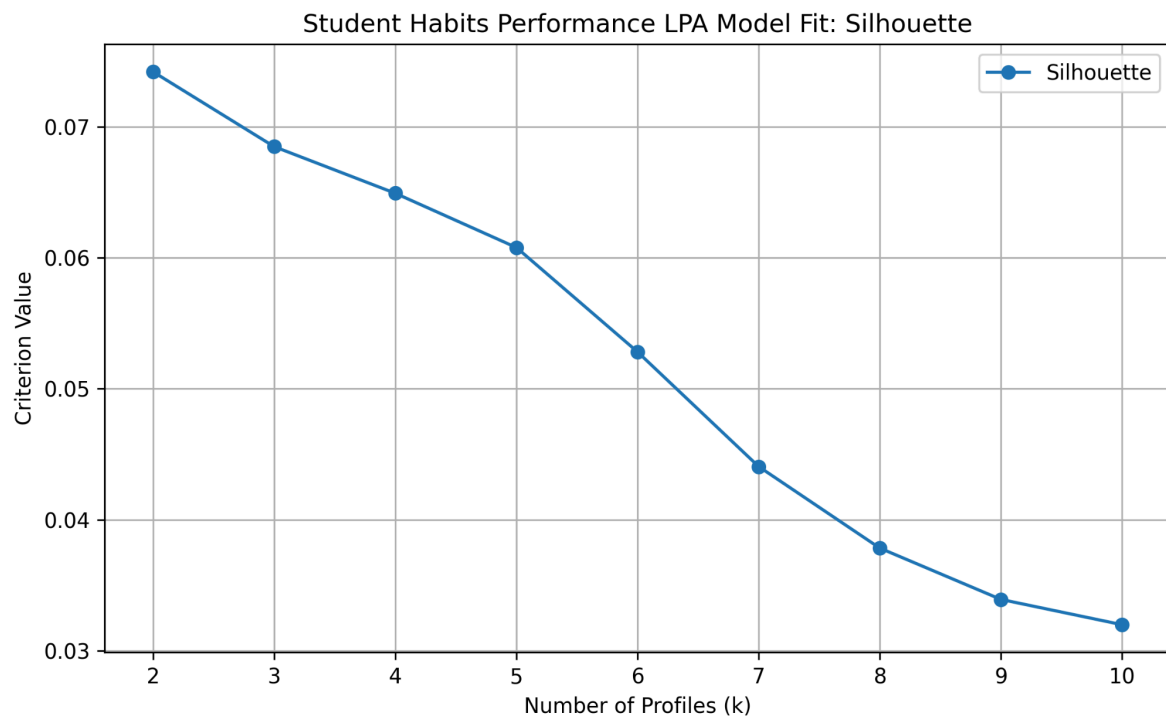


Figure 4.19: BIC & AIC Plot for LPA Evaluation for 'UI/UX Across Popular Digital Platforms'.

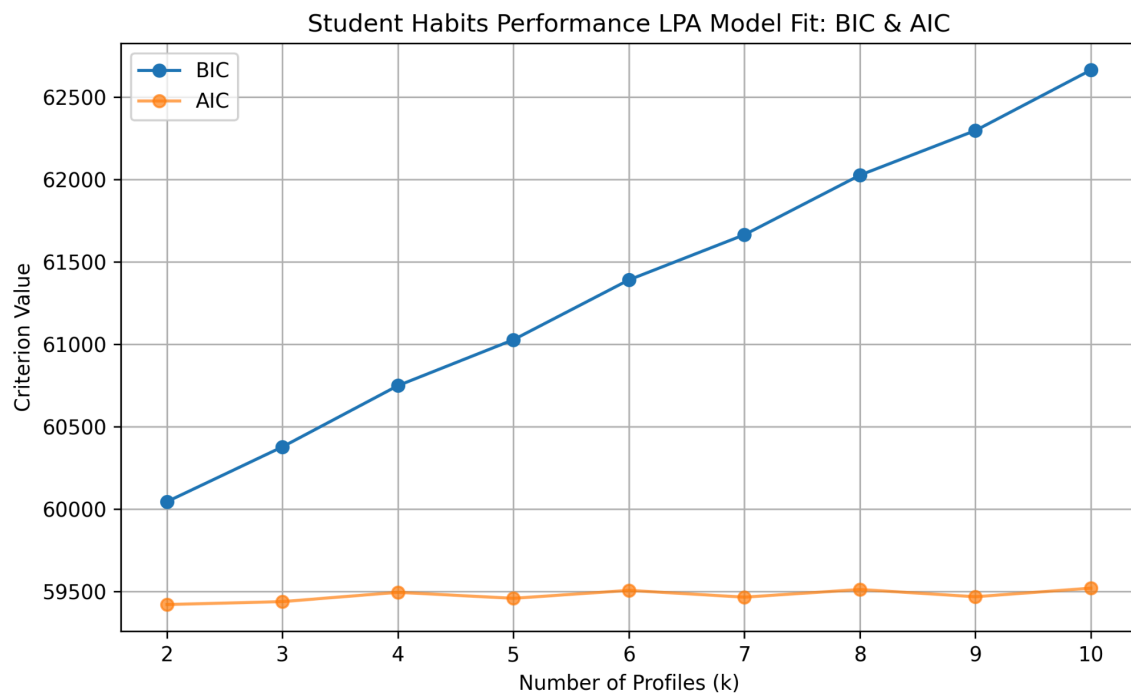


Figure 5.3: K-means Exam Score Distribution plot at $k=4$ for 'Student Habits vs Academic Performance' dataset.

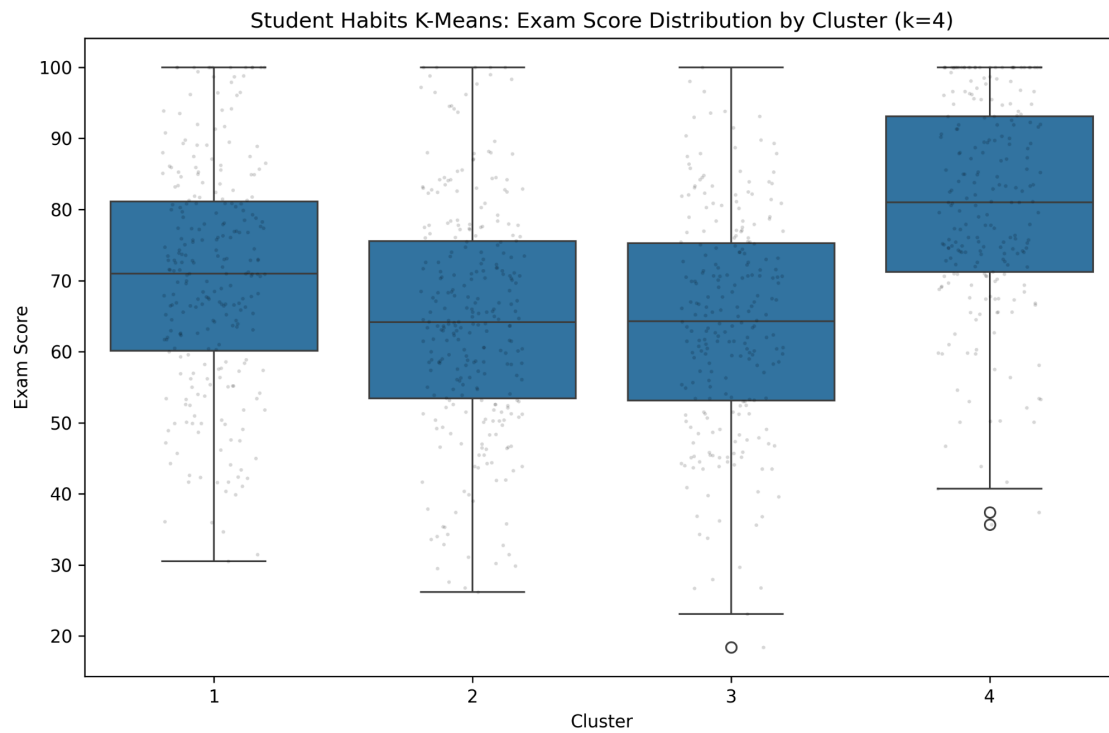


Figure 5.6: Agglomerative Hierarchical Exam Score Distribution plot at $k=4$ for 'Student Habits vs Academic Performance' dataset.

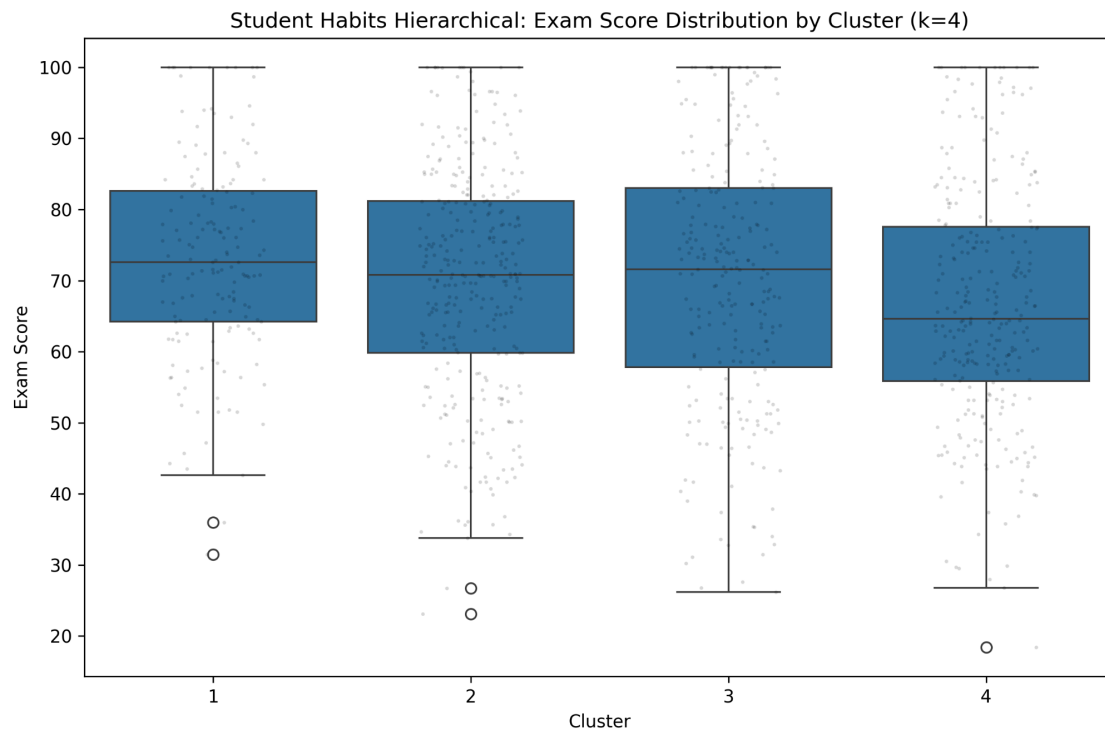


Figure 5.9: Latent Profile Analysis Exam Score Distribution plot at $k=8$ for ‘Student Habits vs Academic Performance’ dataset.

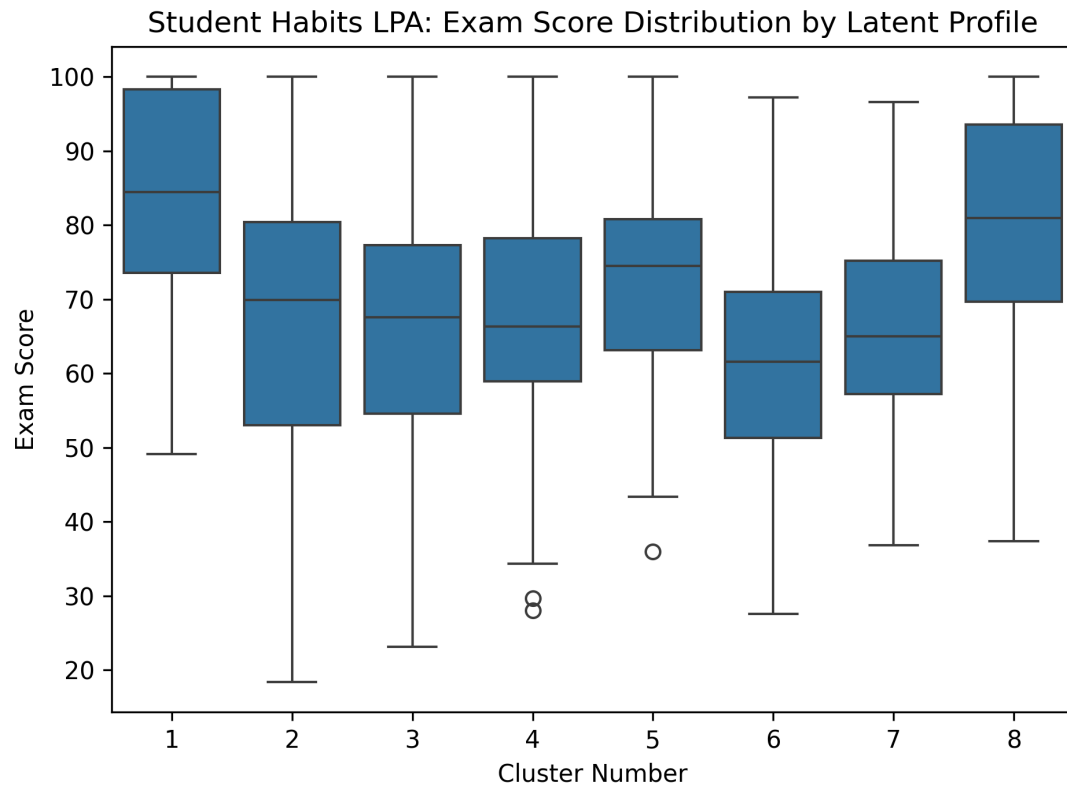


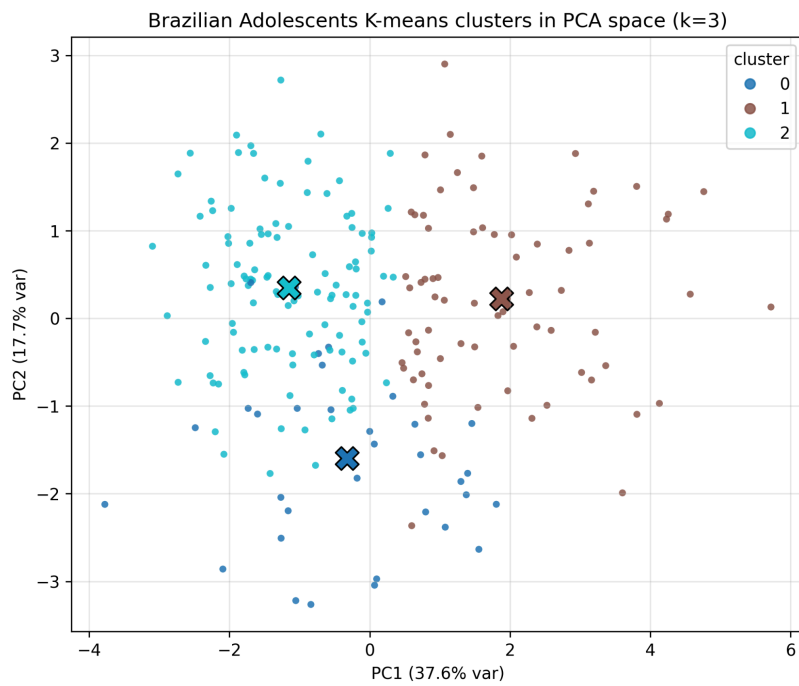
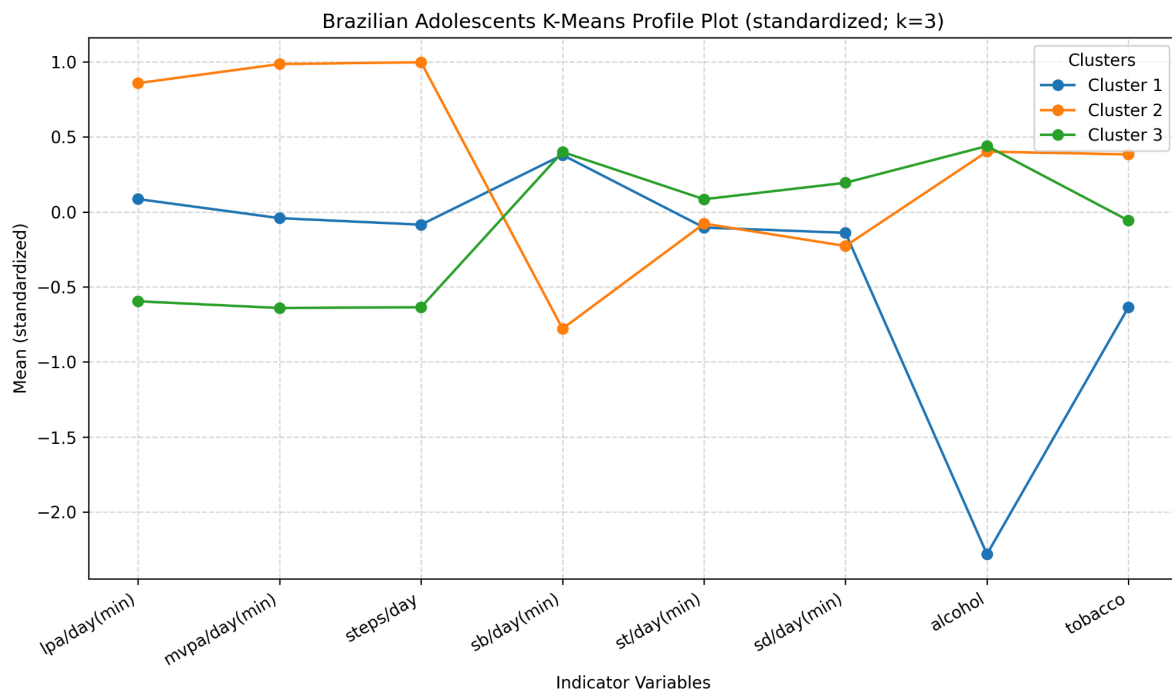
Figure 5.10: K-means PCA plot at $k=3$ for ‘Behavioural Classes Brazilian Adolescents’ dataset.Figure 5.11: K-means Profile line plot at $k=3$ for ‘Behavioural Classes Brazilian Adolescents’ dataset.

Figure 5.12: Agglomerative Hierarchical PCA plot at $k=3$ for ‘Behavioural Classes Brazilian Adolescents’ dataset.

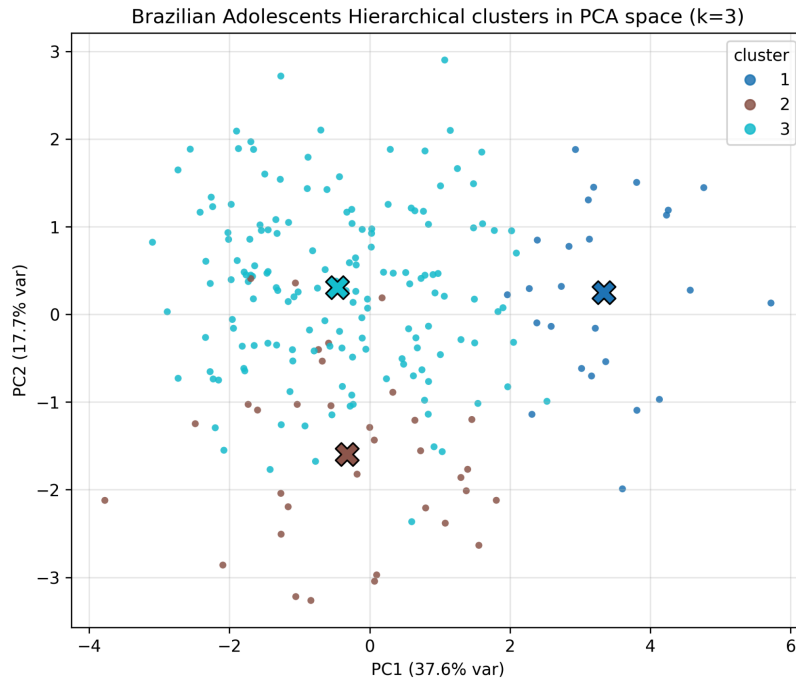


Figure 5.13: Agglomerative Hierarchical Profile line plot at $k=3$ for ‘Behavioural Classes Brazilian Adolescents’ dataset.

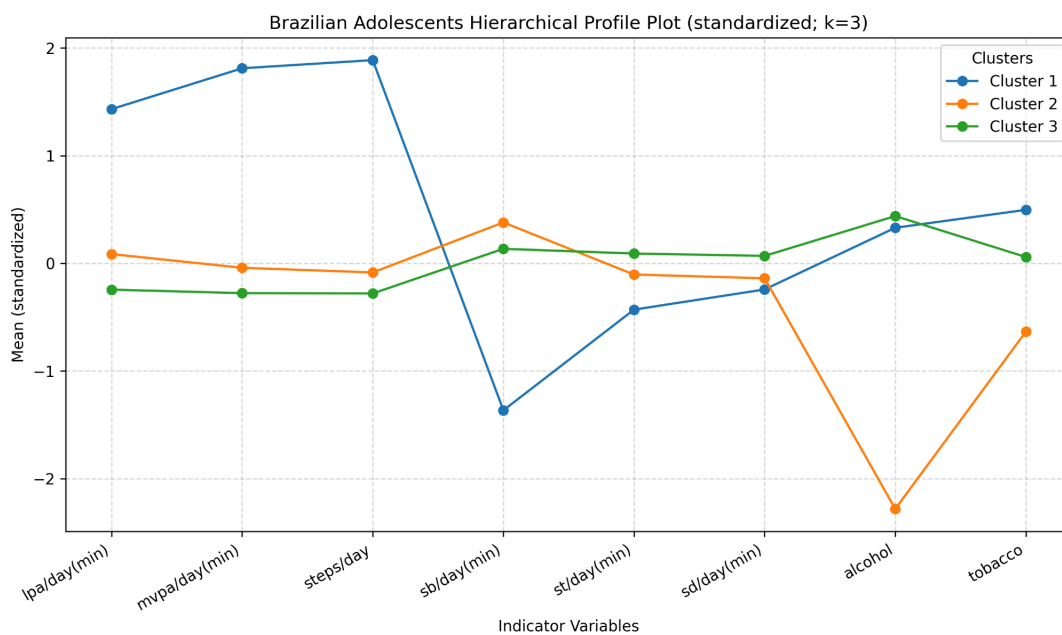


Figure 5.14: Latent Profile Analysis PCA plot at $k=6$ for ‘Behavioural Classes Brazilian Adolescents’ dataset.

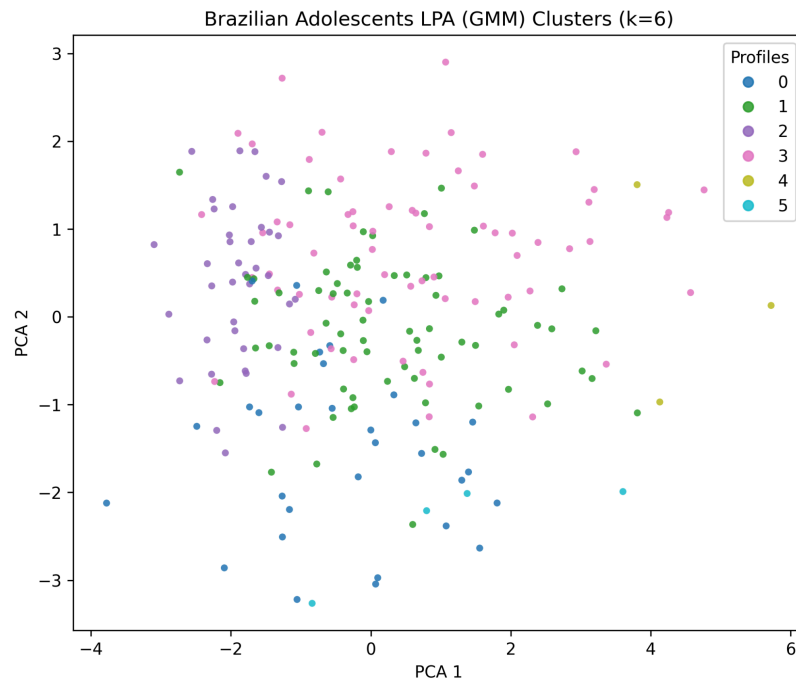


Figure 5.15: Latent Profile Analysis Profile line plot at $k=6$ for “Behavioural Classes Brazilian Adolescents” dataset.

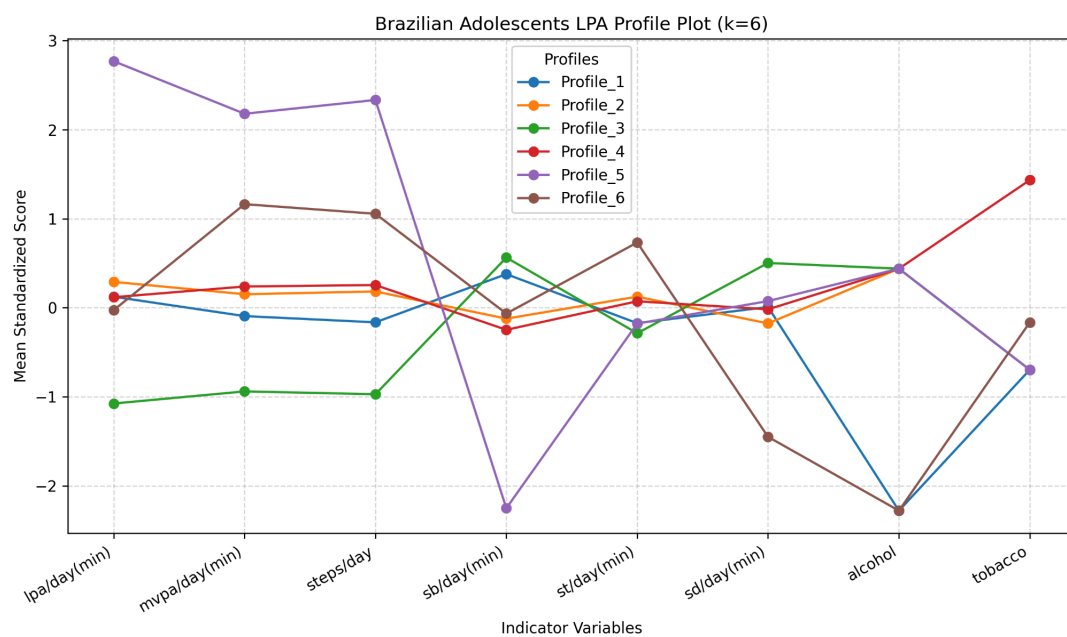


Figure 5.16: K-means PCA plot at $k=4$ for ‘UI/UX User Interaction Across Popular Digital Platforms’ dataset.

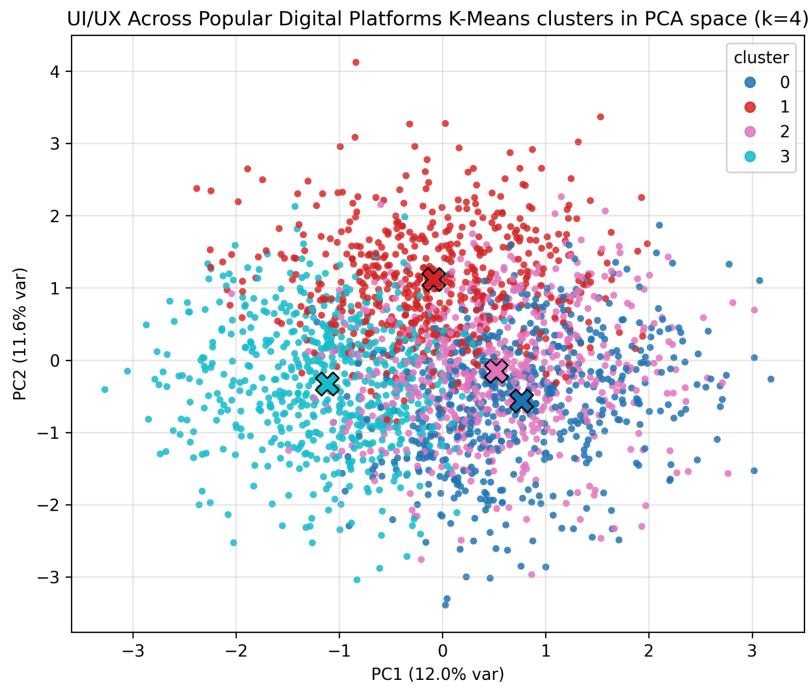


Figure 5.17: K-means Profile Line plot at $k=4$ for ‘UI/UX User Interaction Across Popular Digital Platforms’ dataset.

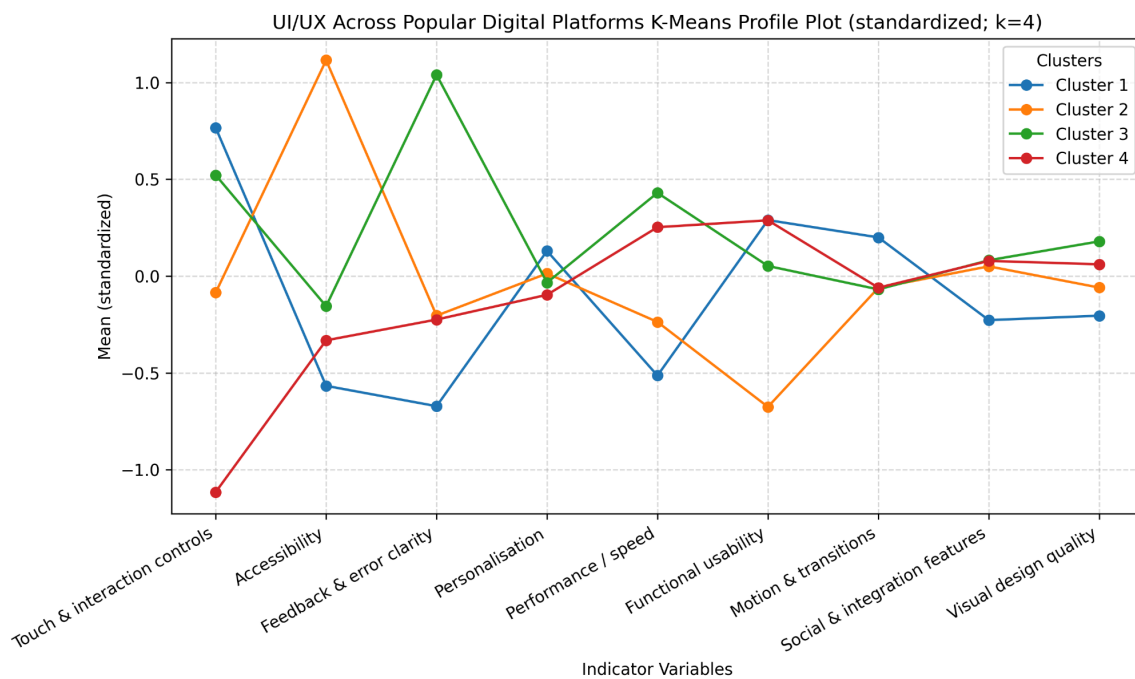


Figure 5.18: Agglomerative Hierarchical PCA plot at $k=4$ for ‘UI/UX User Interaction Across Popular Digital Platforms’ dataset.

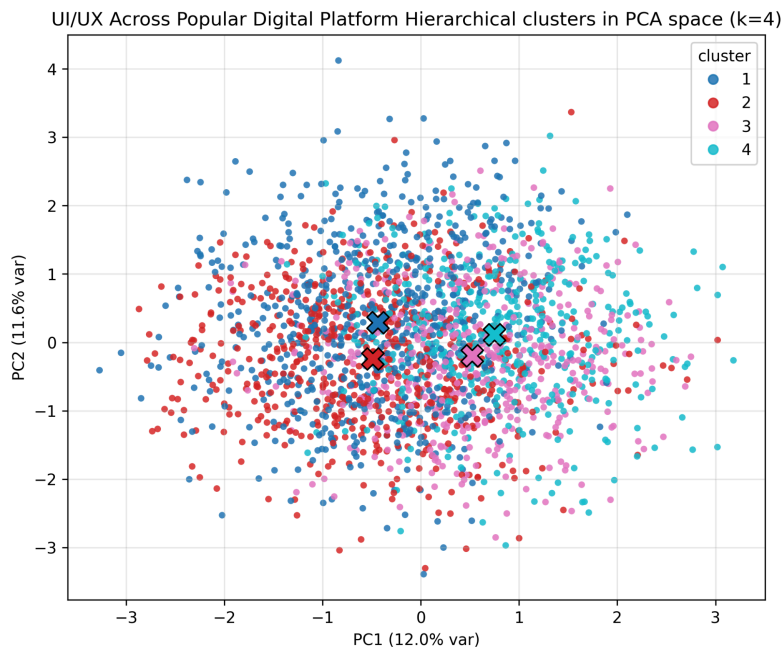


Figure 5.19: Agglomerative Hierarchical Profile Line plot at $k=4$ for ‘UI/UX User Interaction Across Popular Digital Platforms’ dataset.

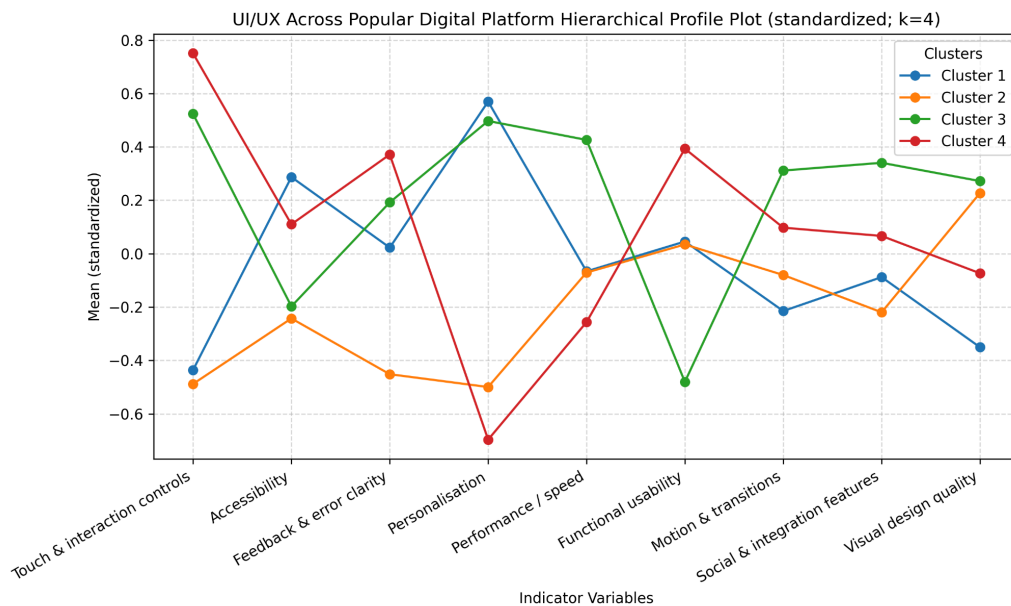


Figure 5.20: Latent Profile Analysis PCA plot at $k=6$ for ‘UI/UX User Interaction Across Popular Digital Platforms’ dataset.

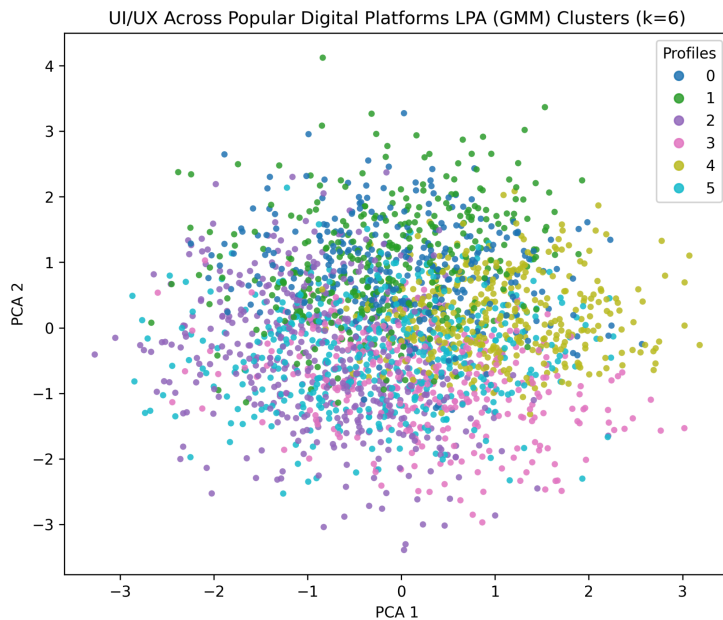


Figure 5.22: LPA non-factorised Profile Line plot at $k=6$ for ‘UI/UX User Interaction Across Popular Digital Platforms’ dataset.

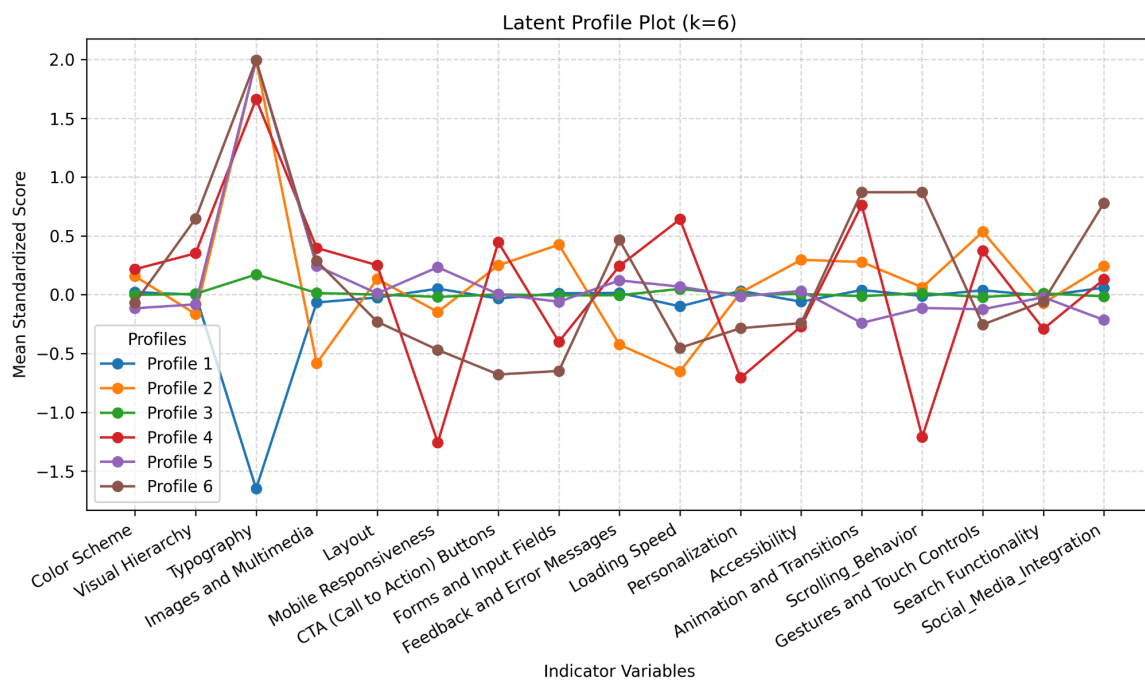


Table 5.4 Behavioural variable summary of cluster profiles for k-means clustering at $k=4$ for “Student Habits vs Academic Performance” dataset.

Cluster #	Cluster Size	Cluster Percentage	Summary	Mean Exam Results
Cluster 1	259	25.9 %	- Lower engagement across study and entertainment. - Lower attendance. - Strong mental health. - Moderate academic performance.	70.63
Cluster 2	276	27.6%	- Very high exercise levels. - Slightly higher entertainment (Netflix). - Lower study time. - Lower mental health. - Lower academic performance.	64.53
Cluster 3	240	24%	- Average study, but very low exercise. - Low entertainment. - Very low mental health. - Slightly higher attendance but still low performance.	64.09
Cluster 4	225	22.5%	- Highest study time. - High social and entertainment use. - Highest attendance. - Good sleep and well-being. - Highest academic performance.	80.51

Table 5.5: Behavioural variable summary of cluster profiles for agglomerative hierarchical clustering at $k=4$ for “Student Habits vs Academic Performance” dataset.

Cluster #	Cluster Size	Cluster Percentage	Summary
Cluster 1	159	15.9%	- Lowest social media and Netflix use. - Lowest attendance. - Slightly below average study time. - Highest sleep levels. - Average exercise and mental health.
Cluster 2	315	31.5%	- Moderate study, social media, and Netflix use. - Above-average attendance. - Average sleep. - Very low exercise levels. - Slightly above-average mental health.
Cluster 3	246	24.6%	- Higher social media use. - Moderate study and Netflix use. - Above-average attendance. - Average sleep. - Highest exercise levels. - Slightly above-average mental health.
Cluster 4	280	28%	- Slightly above-average study time. - Moderate social media and Netflix use. - Average attendance. - Lowest sleep levels. - Average exercise. - Low mental health.

Table 5.6: Behavioural variable summary of cluster profiles for k-means clustering at $k=8$ for “Student Habits vs Academic Performance” dataset.

Cluster #	Cluster Size	Cluster Percentage	Summary
Cluster 1	96	9.6%	- Above-average study time. - Very low social media and Netflix use. - Slightly below-average attendance. - Above-average sleep. - Highest exercise levels. - Average mental health.
Cluster 2	186	18.6%	- Slightly below-average study time. - Average social media and Netflix use. - Above-average attendance. - Slightly above-average sleep. - Lower exercise. - Slightly above-average mental health.
Cluster 3	144	14.4%	- Slightly above-average study time. - Near-average social media and Netflix use. - Average attendance and sleep. - Very low exercise. - Slightly below-average mental health.
Cluster 4	126	12.6%	- Above-average study time. - Slightly above-average social media use. - Slightly below-average Netflix use. - High attendance. - Below-average sleep. - Lower exercise. - Low mental health.
Cluster 5	117	11.7%	- Slightly below-average study time. - Highest social media use. - Near-average Netflix use. - Slightly below-average attendance. - Below-average sleep. - Average exercise. - Highest mental health.
Cluster 6	172	17.2%	- Below-average study time. - Slightly above-average social media and Netflix use. - Slightly below-average attendance. - Slightly below-average sleep. - High exercise. - Low mental health.
Cluster 7	55	5.5%	- Average study time.

			- Slightly lower social media and Netflix use. (2nd lowest screen time) - Lowest attendance. - Above-average sleep. (2nd highest) - Lower exercise. (2nd lowest) - Low mental health. (tied for 2nd lowest)
Cluster 8	104	10.4%	- Above-average study time. - Slightly below-average social media use. - Highest Netflix use. - Above-average attendance. - Average sleep. - High exercise. - High mental health.